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Drone and Satellite-Based Remote Sensing for Early Wheat Disease Detection: A Literature Review

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Abstract

Wheat is one of the most important food crops worldwide, but its production is often threatened by various diseases that can cause serious yield losses. Traditional methods for detecting and managing these diseases are usually slow, rely on manual inspection, and may miss early warning signs. New technologies like remote sensing and artificial intelligence (AI) offer faster, more accurate, and scalable solutions for early disease detection.

This report explores the use of advanced imaging technologies, such as RGB, thermal, multispectral, and hyperspectral cameras along with remote sensing platforms like satellites, airplanes, and drones for monitoring crop health. It also introduces deep learning and its key applications in agriculture, including image classification, object detection, and image segmentation, which allow for automatic and precise identification of wheat diseases.

Special attention is given to the integration of remote sensing data with machine learning and deep learning methods to detect wheat diseases at different growth stages. The report explains the full workflow, from image collection and preprocessing to feature extraction and classification. It also examines how combining data from multiple sources, like satellites and UAVs, can improve accuracy, while discussing the challenges and future opportunities of this approach. These advanced techniques highlight the potential of AI-driven remote sensing to enhance wheat disease management and promote more efficient, sustainable agriculture.

Keywords — Remote sensing, Precision agriculture, Machine learning (ML), Deep learning (DL), Vegetation indices (NDVI, NDRE).

Résumé

Le blé est l'une des cultures vivrières les plus importantes au monde, mais sa production est souvent menacée par diverses maladies pouvant entraîner de lourdes pertes de rendement. Les méthodes traditionnelles de détection et de gestion de ces maladies sont généralement lentes, reposent sur une inspection manuelle et peuvent passer à côté des signes précoces. Les nouvelles technologies, comme la télédétection et l'intelligence artificielle (IA), offrent des solutions plus rapides, plus précises et évolutives pour la détection précoce des maladies.

Ce rapport explore l'utilisation de technologies d'imagerie avancées, telles que les caméras RGB, thermiques, multispectrales et hyperspectrales, ainsi que des plateformes de télédétection comme les satellites, les avions et les drones, pour surveiller l'état de santé des cultures. Il présente également l'apprentissage profond et ses principales applications en agriculture, notamment la classification d'images, la détection d'objets et la segmentation d'images, qui permettent une identification automatique et précise des maladies du blé.

Une attention particulière est accordée à l'intégration des données de télédétection avec des méthodes d'apprentissage automatique et d'apprentissage profond pour détecter les maladies du blé à différents stades de croissance. Le rapport décrit l'ensemble du processus, depuis la collecte et le prétraitement des images jusqu'à l'extraction de caractéristiques et la classification. Il examine aussi comment la combinaison de données issues de multiples sources, telles que les satellites et les drones (UAV), peut améliorer la précision, tout en discutant des défis et des perspectives futures de cette approche. Ces techniques avancées mettent en lumière le potentiel de la télédétection assistée par l'IA pour améliorer la gestion des maladies du blé et promouvoir une agriculture plus efficace et durable.

Mots clés — Télédétection, Agriculture de précision, Apprentissage automatique (ML), Apprentissage profond (DL), Indices de végétation (NDVI, NDRE).

ملخص

يُعد القمح من أهم المحاصيل الغذائية في العالم، لكن إنتاجه غالبًا ما يكون مهددًا بأمراض متعددة يمكن أن تتسبب في خسائر كبيرة في الغلة. الطرق التقليدية للكشف عن هذه الأمراض وإدارتها عادةً ما تكون بطيئة، وتعتمد على الفحص اليدوي، وقد تفشل في اكتشاف العلامات المبكرة. تقدم التقنيات الحديثة مثل الاستشعار عن بُعد والذكاء الاصطناعي حلولاً أسرع وأكثر دقة وقابلة للتوسع للكشف المبكر عن الأمراض.

يستعرض هذا التقرير استخدام تقنيات التصوير المتقدمة، مثل الكاميرات RGB ، والحرارية، والمتعددة الأطياف، وفائقة الطيف، إلى جانب منصات الاستشعار عن بُعد مثل الأقمار الصناعية والطائرات بدون طيار (الدرون)، لمراقبة صحة المحاصيل. كما يقدم التعلم العميق وتطبيقاته الرئيسية في الزراعة، بما في ذلك تصنيف الصور، واكتشاف الأجسام، وتقسيم الصور، والتي تتيح التعرف التلقائي والدقيق على أمراض القمح.

يُولى اهتمام خاص لدمج بيانات الاستشعار عن بُعد مع تقنيات التعلم الآلي والتعلم العميق من أجل الكشف عن أمراض القمح في مراحل النمو المختلفة. يشرح التقرير سير العمل الكامل، بدءًا من جمع الصور ومعالجتها، إلى استخراج الميزات والتصنيف. كما يتناول كيف يمكن أن يؤدي الجمع بين البيانات من مصادر متعددة، مثل الأقمار الصناعية والطائرات بدون طيار، إلى تحسين الدقة، مع مناقشة التحديات والفرص المستقبلية لهذا النهج. تسلط هذه التقنيات المتقدمة الضوء على إمكانات الاستشعار عن بُعد المدعوم بالذكاء الاصطناعي في تحسين إدارة أمراض القمح وتعزيز الزراعة الأكثر كفاءة واستدامة.

الكلمات المفتاحية الاستشعار عن بُعد، الزراعة الدقيقة، التعلم الآلي، التعلم العميق، مؤشرات الغطاء النباتي.

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Abbreviations

AF	**A**ctivation **F**unction
ANN	**A**rtificial **N**eural **N**etwork
BPNN	**B**ack**P**ropagation **N**eural **N**etwork
CNN	**C**onvolutional **N**eural **N**etwork
DL	**D**eep **L**earning
DNN	**D**eep **N**eural **N**etwork
FAO	**F**ood and **A**griculture **O**rganization
GLCM	**G**ray-**L**evel **C**o-**O**ccurrence **M**atrix
GPS	**G**lobal **P**ositioning **S**ystem
LiDAR	**L**ight **D**etection **A**nd **R**anging
ML	**M**achine **L**earning
MLP	**M**ulti-**L**ayer **P**erceptron
NIR	**N**ear-**I**nfrared
PCA	**P**rincipal **C**omponent **A**nalysis
PLSR	**P**artial **L**east **S**quares **R**egression
PSPNet	**P**yramid **S**cene **P**arsing **N**etwork
ResNet	**R**esidual **N**etwork
RGB	**R**ed, **G**reen, **B**lue
SSD	**S**ingle **S**hot **M**ultibox **D**etector
SVM	**S**upport **V**ector **M**achine
SVR	**S**upport **V**ector **R**egression
TF	**T**exture **F**eature
UAV	**U**nmanned **A**erial **V**ehicle
VGGNet	**V**isual **G**eometry **G**roup **N**etwork

VI	**V**egetation **I**ndex
YOLO	**Y**ou **O**nly **L**ook **O**nce

General introduction

Wheat is one of the most widely cultivated and consumed staple crops in the world, playing a critical role in global food security. However, wheat production is increasingly threatened by a variety of biotic factors, most notably diseases that can cause significant yield losses and degrade grain quality. Traditionally, disease detection and control in agriculture have relied on manual inspection, expert intervention, and chemical treatments. These methods are often time-consuming, labor-intensive, and not scalable for large-scale farming operations.

The increasing global demand for food, combined with environmental constraints and the limitations of conventional agricultural practices, has led to the rise of smart agriculture, an approach that leverages advanced technologies to enhance decision-making and optimize farming operations. Within this context, image-based plant disease detection has become an essential tool for early diagnosis and effective crop management. In recent years, deep learning (DL) especially techniques in computer vision, has gained popularity for automating the detection and classification of plant diseases with high precision.

Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated superior performance over traditional machine learning methods in image recognition tasks. These models are capable of automatically learning complex features from raw data, eliminating the need for manual feature extraction. As a result, CNNs and their variants have been widely used in agricultural applications such as disease classification, object detection in field images, and leaf segmentation.

Despite their impressive results, deep learning models often require large labeled datasets and significant computational resources. Additionally, many models are developed under controlled conditions, which limits their applicability in real-world agricultural settings. Challenges such as variable lighting, background noise, and early-stage disease symptoms require robust and adaptable detection systems (Alzubaidi et al., 2021).

To address these challenges, recent advances have focused on the integration of remote sensing technologies and AI-based models. Remote sensing allows for large-scale monitoring by capturing spectral, thermal, and spatial information from various platforms including satellites, drones (UAVs), and multispectral or hyperspectral cameras. When combined with AI and deep learning, remote sensing data enables early detection, classification, and spatial mapping of crop diseases with greater accuracy and efficiency (Wang et al., 2024).

This report explores the synergy between AI-based methods and remote sensing technologies for the detection of wheat diseases. It provides a comprehensive overview of the current challenges in wheat disease management and highlights modern approaches to tackle them using artificial intelligence.

The structure of the report is as follows:

- **Chapter 1**: Introduces remote sensing technologies and platforms used in agricultural monitoring. It covers various imaging technologies (e.g., RGB, NIR, thermal, multispectral, LiDAR, hyperspectral), sensing platforms (satellites, aircraft, UAVs), vegetation indices, image processing techniques, and applications of remote sensing data in agriculture.
- **Chapter 2**: Explores the fundamentals of deep learning with a focus on its application in visual recognition tasks for wheat disease detection. It discusses deep neural networks, training and optimization strategies, evaluation methods, and core computer vision tasks such as image classification, object detection, and segmentation.
- **Chapter 3**: Integrates the knowledge from previous chapters to examine how remote sensing and AI-based methods can be combined for the early detection of wheat diseases. It details wheat growth stages, common diseases, the workflow for ML/DL-based detection using remote sensing data, challenges in integration, fusion of UAV and satellite data, and outlines future perspectives.

Chapter 1

Remote Sensing Technologies and Platforms for Agricultural Monitoring

1.1 Introduction

Remote sensing has become a key technology in modern agriculture, enabling efficient crop monitoring through aerial and satellite imagery. It allows for non-invasive observation of large agricultural areas, supporting early detection of stress and disease.

This chapter introduces the main imaging technologies such as RGB, multispectral, and hyperspectral cameras. It also covers the platforms used for data collection, including satellites, aircraft, and UAVs. Vegetation indices and basic image processing techniques are explored to understand plant health. Lastly, practical applications of remote sensing in agriculture are highlighted.

1.2 Imaging technologies in remote sensing

Imaging technologies play a key role in remote sensing systems by capturing high-resolution images of the Earth's surface, which are crucial for agricultural monitoring. These technologies use different types of cameras to detect various parts of the electromagnetic spectrum (Figure 1.1), enabling detailed analysis of vegetation, soil, and environmental conditions. The choice of camera depends on the specific agricultural purpose, such as disease detection, crop monitoring, or environmental assessment.

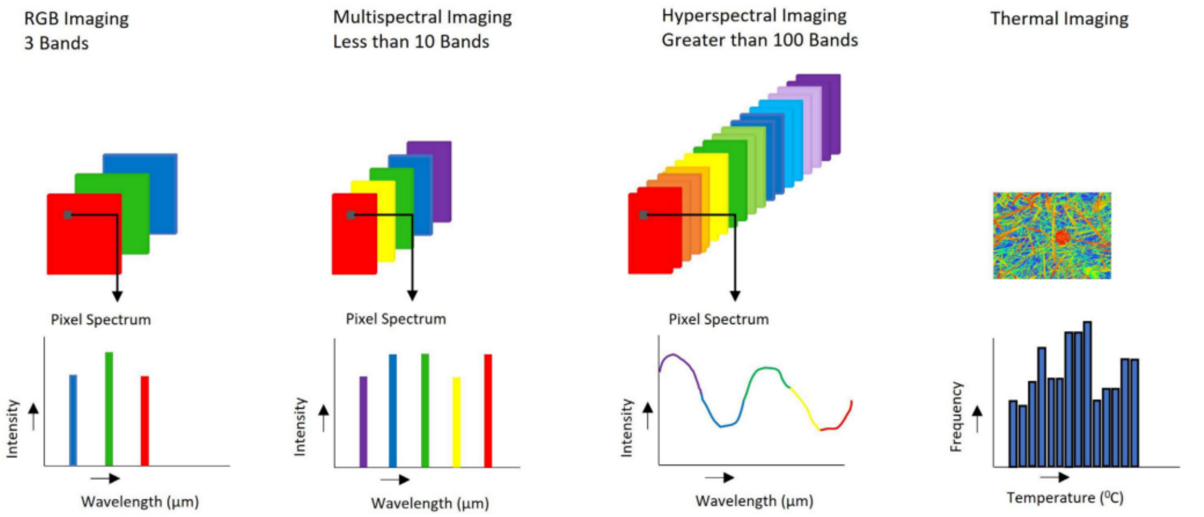

Figure 1.1: Image acquisition techniques (Ghazal et al., 2024).

1.2.1 RGB cameras (Red, Green, Blue)

RGB cameras are the most common type of imaging technology used in remote sensing. They capture images in the visible light spectrum (red, green, and blue wavelengths), similar to how the human eye perceives the world (Delavarpour et al., 2021).

1.2.2 Near-Infrared (NIR) cameras

Near-Infrared cameras capture images in the near-infrared spectrum, which lies just beyond the visible light range. These cameras are sensitive to light wavelengths that are not visible to the human eye, typically ranging from 700 to 1300 nm (Delavarpour et al., 2021). near-infrared (NIR) imaging is particularly valuable in plant health monitoring as it can detect subtle changes in vegetation that are not visible in the visible light spectrum.

1.2.3 Thermal infrared cameras

Thermal infrared cameras capture images based on the heat emitted by objects, operating in the infrared spectrum (wavelengths typically ranging from 8,000 to 14,000 nm). They detect temperature differences and translate them into visual representations, making them invaluable for identifying heat-related patterns in crops and soil (Delavarpour et al., 2021).

1.2.4 Multispectral cameras

Multispectral cameras capture images across multiple spectral bands, including visible (red, green, blue) and non-visible wavelengths (near-infrared and red-edge) across a limited number of discrete bands, each covering a wider spectral range spanning tens to hundreds of nanometers (Lu et al., 2020). These cameras typically have 5 to 10 bands, enabling detailed analysis of plant health and environmental conditions (Delavarpour et al., 2021). They are essential for applications that require data beyond what RGB cameras can provide.

1.2.5 LiDAR cameras (Light Detection and Ranging)

LiDAR cameras use laser pulses to measure distances by calculating the time it takes for the laser to bounce back from a surface. This technology generates high-resolution 3D maps, capturing detailed information about the shape and structure of the terrain, vegetation, and other objects. LiDAR operates effectively in various environmental conditions, including low light or dense canopy cover (Delavarpour et al., 2021).

1.2.6 Hyperspectral cameras

Hyperspectral cameras capture images across a vast range of continuous, narrow spectral bands, often exceeding hundreds. These bands typically have a spectral resolution below 10 nanometers, covering both visible and non-visible wavelengths. The fine-scale spectral data collected allows for the detection of subtle variations in reflectance, enabling detailed analysis of crop conditions, such as disease identification and nutrient stress. This high-resolution imaging is crucial for precision agriculture, where detecting subtle differences can significantly impact decision-making and yield optimization (Delavarpour et al., 2021) (Lu et al., 2020).

1.3 Remote Sensing platforms

Remote sensing platforms are categorized based on their altitude and mobility, each offering unique capabilities for capturing agricultural data. The three main types are satellite-based imaging, UAV (drone) imaging, and aircraft imaging.

1.3.1 Satellite-Based imaging

Satellite-based imaging involves the use of Earth-observing satellites equipped with multispectral sensors to capture detailed images of the Earth's surface. These satellites, such as Quickbird and Landsat-1, can capture high-resolution images, providing data with varying levels of precision. Early satellite systems offered up to 6.5 meters per pixel in resolution (Phang et al., 2023a), while modern systems have significantly improved both spatial and spectral resolution. These advancements enable more accurate and comprehensive monitoring of large areas, laying the foundation for a wide range of applications, including agriculture and environmental studies (Adão et al., 2017). Despite its many advantages, satellite-based imaging also presents several challenges and limitations that must be considered in agricultural applications, including:

- **Coarse Resolution:** The resolution of satellite imagery can be too coarse for some applications. For instance, imagery from platforms like Sentinel-2 (up to 10 m resolution) may not be detailed enough for fields with closely spaced rows, such as vines, leading to mixed pixel data that combines multiple rows and soil (Adão et al., 2017).
- **Data Processing Complexity:** Due to the coarse resolution, advanced methods like computer vision classifiers and statistical decision trees are often required to extract useful information, such as detecting shrubs or distinguishing plant species (Phang et al., 2023b).

- **Mixed Pixel Issue:** Low-resolution pixels, like those from Landsat, result in mixed data, complicating analysis and interpretation, especially in detailed applications (Delavarpour et al., 2021).
- **Spatiotemporal Challenges:** Obtaining timely spatiotemporal data on crop phenological status during critical growth periods is difficult, especially due to cloud coverage (Delavarpour et al., 2021).
- **High Costs:** Accessing satellite data equipped with multispectral sensors can be expensive, which limits its use for some applications (Delavarpour et al., 2021).

1.3.2 Aircraft-Based imaging

Aircraft-based imaging involves the use of piloted or unpiloted aircraft, such as airplanes, equipped with various remote sensing tools to capture high-resolution imagery and sensor data over large agricultural areas. These systems offer a significant advantage in large-scale applications due to their flexibility and ability to carry heavier payloads of sensors, making them a viable alternative to satellite-based or UAV-based solutions (Delavarpour et al., 2021). Before the widespread adoption of UAVs, manned aircraft were commonly employed for lower-to-ground remote sensing, utilizing multi-spectral or electro-optic (EO) sensors to monitor agricultural conditions (Phang et al., 2023a), However, this approach also faces several limitations, including:

- **High Operational Costs:** Operating aircraft-based systems involves significant expenses for fuel, maintenance, and pilot salaries, making them less affordable compared to UAVs (Delavarpour et al., 2021; Phang et al., 2023b).
- **Complex Logistics and Pilot Requirement:** Certified pilots and specific flight logistics are essential, adding complexity and reducing flexibility in deployment (Adão et al., 2017).
- **Limited Flexibility:** Aircraft require designated takeoff and landing zones, restricting their use in remote or uneven terrains (Delavarpour et al., 2021).
- **Weather Sensitivity:** Adverse weather conditions, such as strong winds or rain, can impact flight stability and data quality, limiting operations during critical periods (Delavarpour et al., 2021).
- **Regulatory and Airspace Restrictions:** Aircraft-based systems are subject to strict aviation regulations and airspace restrictions, limiting their operational scope (Delavarpour et al., 2021).
- **High Data Processing Requirements:** Large volumes of data generated require specialized software and computing resources, making data processing resource-intensive (Delavarpour et al., 2021).
- **Costly for Large-Scale Use:** The expense and complexity of these systems make them impractical for frequent large-scale monitoring, favoring UAV alternatives for cost efficiency (Phang et al., 2023b).
- **Resolution Limitations:** Although aircraft provide better resolution than satellites, their imagery can still be too coarse for some precision applications (Adão et al., 2017).

1.3.3 UAV-Based imaging

Unmanned Aerial Vehicles (UAVs), or drones, are versatile remote sensing platforms that have become essential tools in remote sensing due to their cost-effectiveness, flexibility, and ability to capture high-resolution (cm-level) images, making them ideal for precision agriculture (Sishodia et al., 2020). UAV-based imaging typically involves using low-altitude remote Sensing Systems to acquire detailed imagery of the Earth's surface at low altitudes, providing high precision and adaptability (Phang et al., 2023a).

Unlike traditional satellite platforms, UAVs offer several advantages, including on-demand data collection, high-resolution imagery, and flexible deployment, which enable real-time monitoring and analysis (Phang et al., 2023a). A complete Unmanned Aerial System (UAS) includes the UAV and its remote sensing equipment, operating without a human pilot onboard, and is capable of carrying various sensors tailored to different agricultural needs (Adão et al., 2017), Despite their advantages, UAVs also face some challenges, including:

- **Short Flight Duration:** UAVs, especially smaller models, often have flight durations of less than 30 minutes, which limits their coverage area, particularly for large-scale agricultural operations (Phang et al., 2023b).
- **Regulatory Challenges:** Stricter regulations, especially for larger UAVss, slow down their adoption and innovation, hindering their widespread use (Delavarpour et al., 2021; Phang et al., 2023b).
- **Scalability:** UAV-based remote sensing requires trained pilots and continuous monitoring, which limits scalability, particularly for small-scale applications (Phang et al., 2023b).
- **Environmental Sensitivity and Calibration Issues:** Hyperspectral sensors on UAVss face challenges related to environmental factors such as light exposure and atmospheric interference, necessitating frequent recalibration (Adão et al., 2017).
- **Weather Dependency:** UAVs are susceptible to weather conditions like wind and rain, which can affect both flight stability and data quality (Delavarpour et al., 2021).
- **Battery Life:** The limited battery life of UAVss impacts their ability to cover large areas, particularly in extensive agricultural fields (Delavarpour et al., 2021).

1.4 Vegetation indices

Vegetation indices are mathematical formulas that combine light reflectance data from different spectral bands (such as visible, near-infrared, and mid-infrared) captured by sensors on drones or satellites to assess plant health and condition. These indices provide valuable insights into growth stages, plant vigor, biomass, and chlorophyll levels (Sishodia et al., 2020). Plants reflect sunlight differently based on their type, structure, and water content: they reflect little in the blue and red regions, more in green (which makes them appear green), and strongly in the near-infrared (NIR) if healthy (Figure 1.2).

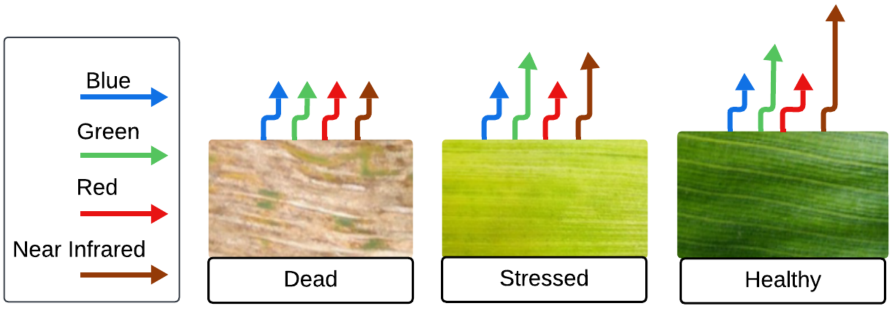

Figure 1.2: Unique optical reflectance signature differences of blue, green, red, and near-infrared light emitted from dead, stressed, and healthy plant tissue (Olson & Anderson, 2021).

Thermal and infrared indices, including CWSI (Crop Water Stress Index) and SIWSI (Short-wave Infrared Water Stress Index), use thermal radiation data to estimate plant temperature, water stress, and transpiration, aiding in drought detection, crop yield estimation, and disease monitoring (Sishodia et al., 2020). For more vegetation indices, their mathematical formulas, and detailed applications, refer to Appendix 3.9.

1.5 Image processing

Imagery processing is the technique used to turn multiple aerial images captured by drones (UAVs) or remote sensors into a single, accurate, and clear image known as an orthomosaic. Since drones don't capture one large image but instead take many smaller ones of different parts of the area, these images must be merged using a process called image stitching (as shown in Figure 1.3), This involves merging individual images into one large composite by identifying and aligning "key points" such as a rock, plant, or edge of a field. While the drone captures these images, it simultaneously records global positioning system (GPS) or location metadata, which is then used in geographic alignment to position each image on a map accurately. The final output of this process is known as data products, which can include color mosaics (stitched colored images), spectral mosaics (capturing invisible wavelengths like near-infrared for agricultural analysis), thermal mosaics (highlighting temperature variations), surface and terrain models (representing elevation and landscape shapes), and point clouds (dense 3D representations of surfaces for detailed analysis) (Olson & Anderson, 2021).

Table 1.1: Some recently used vegetation indices for remote sensing applications in precision agriculture (Sishodia et al., 2020).
Note: *Red, Green, and Blue* refer to visible bands of the electromagnetic spectrum. *NIR (Near-Infrared)* refers to wavelengths (700–1300 nm). *Red Edge* is the narrow spectral region (680–750 nm) between the red and NIR bands.

Spectral Index	**Equation**	**Application**
Normalized Difference Vegetation Index (NDVI)	$NDVI = \frac{\text{NIR}-\text{Red}}{\text{NIR}+\text{Red}}$	Measure of healthy, green vegetation
Green Normalized Difference Vegetation Index (GNDVI)	$GNDVI = \frac{\text{NIR}-\text{Green}}{\text{NIR}+\text{Green}}$	Measure of healthy, green vegetation; higher chlorophyll concentration sensitivity than NDVI
Red Edge Normalized Difference Vegetation Index (RENDVI)	$RENDVI = \frac{\text{NIR}-\text{Red Edge}}{\text{NIR}+\text{Red Edge}}$	Modification of NDVI; utilizes the red edge for enhanced sensitivity to vegetation stress
Soil Adjusted Vegetation Index (SAVI)	$SAVI = \frac{1.5(\text{NIR}-\text{Red})}{\text{NIR}+\text{Red}+0.5}$	Modification of NDVI; suppresses the effects of soil pixels
Atmospherically Resistant Vegetation Index (ARVI)	$ARVI = \frac{\text{NIR}-(\text{Red}-\text{Blue})}{\text{NIR}+(\text{Red}-\text{Blue})}$	Designed to be less affected by atmospheric conditions; used for disease and weed mapping
Wide Dynamic Range Vegetation Index (WDRVI)	$WDRVI = \frac{\text{NIR}-\text{Red}}{\text{NIR}+\text{Red}}$	N-Application, yield estimation, crop growth (LAI), and disease detection
Normalized Difference Red Edge Index (NDRE)	$NDRE = \frac{\text{NIR}-\text{Red Edge}}{\text{NIR}+\text{Red Edge}}$	Used for crop yield estimation, biomass, nitrogen management, and disease detection
Red Edge Difference Vegetation Index (REDVI)	$REDVI = \text{NIR} - \text{Red Edge}$	Estimates biomass, nitrogen uptake, and nutrient concentration

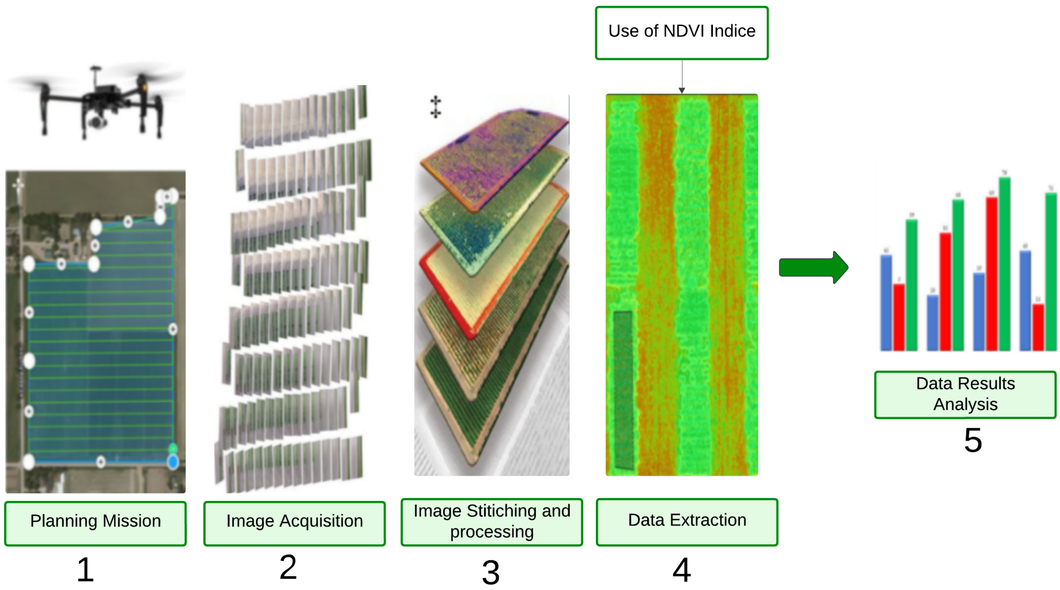

Figure 1.3: Standard Workflow for Generating UAV-Derived Imagery and Data Products (Olson & Anderson, 2021).

Imagery can be processed on-site with personal computers and the appropriate software or in the field with cloud computing imagery processing (Olson & Anderson, 2021). The pros and cons of each imagery processing option are discussed in Table 1.2.

	Pros	**Cons**
Software (On-Site Processing on PC)	Customizable options	High-performance PC needed
	No limits on images	Expensive initial cost
	GIS software compatible	Steep learning curve
	One-time purchase	Slower processing
Cloud Computing (Online/Remote Processing)	Easy to use	Fewer customizations
	Low upfront cost	Image limits
	Fast field processing	Ongoing costs
	No powerful PC needed	
	Simplified analysis	

Table 1.2: Comparison of On-Site Software vs. Cloud Computing for Imagery Processing (Olson & Anderson, 2021).

1.6 Applications of remote sensing data

Remote sensing has diverse applications in agriculture, enabling precise, data-driven decision-making. It helps monitor field conditions, optimize resource use, and address issues early. The following sections highlight key areas where remote sensing enhances farm management:

- **Crop Health Monitoring:** Remote sensing enables regular monitoring of crop conditions, helping detect diseases, pest presence, and water or nutrient stress early. Vegetation indices (e.g., NDVI) derived from RGB, NIR, multi-spectral, and hyperspectral sensors are commonly used to assess aspects such as foliage cover, pigment composition, and water stress. LiDAR and GPS can also support the accurate mapping of plant structure and spatial positioning. These technologies collectively support better crop management and yield optimization (Phang et al., 2023a).

- **Weed Control:** Precise weed detection helps reduce competition for water, nutrients, and light. Aerial imagery from UAVs and satellite platforms, often using RGB and multispectral cameras, supports vegetation indexing and mapping for weed identification. Some systems enable real-time onboard image analysis and spraying. These tools assist in implementing targeted and efficient weed management strategies while reducing input costs (Phang et al., 2023a).

- **Infectious Disease Epidemiology and Mapping:** UAVs are used to gather high-resolution spatial data for studying the relationships between environmental conditions and disease spread. They help monitor changes in land use, population distribution, and vegetation patterns, which are valuable for identifying and mitigating disease risks affecting crops or livestock. UAV-collected data provide flexible, cost-effective alternatives to satellite and manned aerial surveys (Phang et al., 2023a).

1.7 Conclusion

In this chapter, we explored the foundations of remote sensing technologies and their crucial role in agricultural monitoring, particularly for detecting plant stress and diseases. We reviewed various imaging systems, data acquisition platforms, and key vegetation indices used to assess crop health. While remote sensing provides rich and valuable data, its full potential is unlocked when combined with intelligent analysis methods. In the next chapter, we shift focus to another technological domain, deep learning and computer vision, which offer powerful tools for automating and enhancing the interpretation of agricultural imagery, especially for disease recognition in wheat crops

Chapter 2

Applications of Deep Learning in Visual Recognition of Wheat Diseases

2.1 Introduction

The spread of wheat diseases poses serious threats to agricultural productivity and food security. Early and accurate detection is essential for timely intervention and disease management. In this context, deep learning has emerged as a powerful approach to support smart agriculture by enabling automated, data-driven solutions.

This chapter focuses on the application of deep learning techniques to visual data, particularly images captured in the field. It highlights three key computer vision tasks used in precision agriculture: image classification, image segmentation, and object detection. These tasks allow machines to recognize disease types, locate affected areas on plant surfaces, and detect objects such as crops or weeds in complex environments. The chapter begins with the fundamentals of deep neural networks and proceeds to how they are used for agricultural image analysis. It also introduces practical techniques like transfer learning and fine-tuning, which help adapt models to specific agricultural datasets.

2.2 Deep learning fundamentals

In this section, we introduce the core principles of DL, focusing on deep neural network (DNN)s and their role in learning complex patterns from data. We explore the structure and components of neural networks, the learning process, optimization techniques, and model evaluation methods. Additionally, we highlight regularization strategies to prevent overfitting.

2.2.1 Deep neural network basics (DNN)

Neural networks form the backbone of deep learning, enabling machines to learn patterns and make predictions from data. Inspired by the structure of the human brain, these networks consist of interconnected layers of artificial neurons that hierarchically process information.

Definition of DNN

Before defining deep neural networks, we first need to understand two essential components:

- **Artificial neuron**: An artificial neuron is the basic building block of artificial neural networks, designed based on the structure and functionality of biological neurons. It receives weighted inputs, processes them through a transfer function, and outputs the result. The artificial neuron model simplifies the biological process where information is received through dendrites, processed in the soma, and transmitted via the axon, as shown in Figure 2.1 (Krenker et al., 2011).
- **Layer**: A layer in a neural network is a set of neurons that perform a specific operation on the data (Bengio et al., 2017).

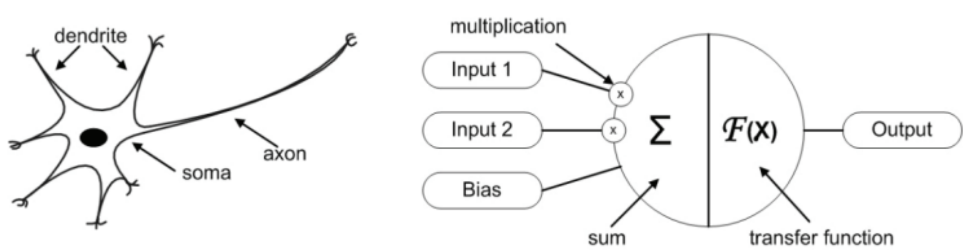

Figure 2.1: Biological Neuron Structure and Its Mathematical Model Representation (Krenker et al., 2011).

By combining multiple layers of interconnected artificial neurons, we arrive at the concept of a Deep Neural Network (DNN). A DNN is a neural network that contains multiple hidden layers between the input and output layers. These additional layers enable the network to learn complex patterns and high-level features from data. Each layer transforms its input into a more abstract representation, improving the network's ability to recognize intricate structures and relationships (A. Li et al., 2021).

Structure of DNN

A neural network (Figure 2.2) consists of three main layers (Ren & Du, 2023):

The input layer: Represents the features of the input data, such as pixel values in an image, denoted as a vector

$$X = [x_1, x_2, \ldots, x_n].$$

The hidden layers: Process this input using weighted connections and biases, computed as:

$$z = W \cdot X + b_z \tag{1}$$

$$F(z) = a \tag{2}$$

where:

- W is the weight matrix,

- b is the bias vector,
- z is the pre-activation value,
- $F(z)$ is the activation function applied to z.

Each neuron in the hidden layers applies an activation function to capture complex patterns.

The output layer: Generates the network's prediction, with the number of neurons corresponding to the specific task, such as one neuron for binary classification or multiple neurons for multiclass classification.

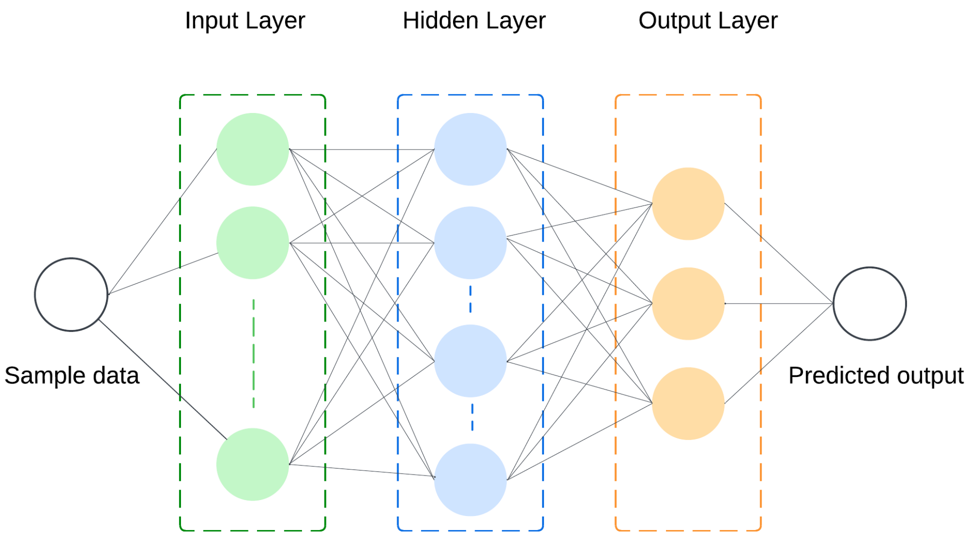

Figure 2.2: The structure of the DNN (Ren & Du, 2023).

Activation functions (AF)

An activation function (AF) is a mathematical function applied to a neuron's output in a neural network to introduce non-linearity. Without an activation function, a neural network with multiple layers would behave like a single-layer perceptron, limiting its ability to model complex relationships (Dubey et al., 2022). Activation functions decide whether a neuron should be activated based on its input. Here are some used activation functions in DNN models (Dubey et al., 2022):

- **Sigmoid function**: Commonly used in binary classification problems, especially in the output layer of models predicting probabilities. It maps input values to a range between 0 and 1, making it suitable for representing probabilities, as shown in equation 2.1.

$$\text{Sigmoid}(x) = \frac{1}{1+e^{-x}} \tag{2.1}$$

- **Tanh (Hyperbolic tangent)**: Often used in hidden layers of neural networks. It outputs values centered around zero, helping reduce bias during optimization. Its formulation is given in equation 2.2.

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{2.2}$$

- **ReLU (Rectified linear unit)**: Widely used in hidden layers due to its simplicity and effectiveness in handling the vanishing gradient problem. It outputs zero for negative inputs and the input itself for positive values, as defined in equation 2.3.

$$\text{ReLU}(x) = \max(0, x) \tag{2.3}$$

- **Softmax function**: Used in the output layer for multi-class classification. It converts a vector of values into a probability distribution, as illustrated in equation 2.4.

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}} \tag{2.4}$$

  where x_i is the current element and the denominator sums over all elements x_j in the vector.

2.2.2 Learning process in deep neural networks

In the information processing flow within an artificial neuron, several elements, known as parameters, are learned from the training data. These parameters include (Kelleher, 2019):

- **Weights**: Weights control the amount of each input feature that passes through the neuron. They represent the coefficients of the connections between neurons in the layers of a neural network. Weights are essential for determining the influence of each input on the output.

- **Biases**: Biases are values added to the outputs of the neurons before applying the activation function. They allow the network to shift the activation function, providing more flexibility to the model.

The learning process in neural networks involves training the model to map input data to desired outputs. This is achieved through the mechanisms of forward propagation and backward propagation, along with optimization techniques that refine the network's parameters (weights and biases) to minimize the error (Kelleher, 2019):

- **Forward Propagation**: In forward propagation, the input data passes through the network layer by layer. Each neuron in a layer performs a weighted sum of its inputs, applies an activation function, and passes the result to the next layer. The process continues until the output layer is reached and a prediction is made.

- **Backward Propagation**: Backward propagation (or backpropagation) is used to update the network's weights. After calculating the error (the difference between the predicted and actual output), the error is propagated back through the network. The weights are adjusted based on the gradient of the error with respect to each weight, using optimization algorithms like gradient descent.

2.2.3 Model training concepts

Effective model training ensures that neural networks learn patterns in data while minimizing errors. This section highlights essential components of the training process.

Loss Functions

Loss functions quantify the error between predicted and actual values, guiding weight updates during optimization (Alzubaidi et al., 2021). Common loss functions include:

- **Cross-Entropy Loss**: Used for classification tasks to measure the divergence between predicted probabilities and true labels.
- **Mean Squared Error (MSE)**: Applied in regression, computing the average squared difference between predicted and actual values.

Overfitting prevention

Overfitting occurs when a model learns noise from training data, reducing generalization to unseen data. Common techniques to mitigate overfitting include:

- **Dropout**: Randomly deactivate neurons during training to enhance robustness.
- **Data Augmentation**: Transform training samples (e.g., rotation, scaling) to increase dataset diversity.

Batch normalization

Batch Normalization normalizes layer outputs to follow a unit Gaussian distribution, a normal distribution with a mean of 0 and a standard deviation of 1, which keeps values in a consistent range and stabilizes learning. This helps stabilize and speed up neural network training by reducing internal covariance shift, which is caused by changing activation distributions during training, especially with diverse data sources. Key benefits include preventing vanishing gradients, improving weight initialization, speeding up convergence, and reducing sensitivity to hyperparameters (Alzubaidi et al., 2021).

2.2.4 Optimization methods and strategies

Optimization techniques in deep learning are methods used to minimize the loss function during training, improving the model's accuracy. These techniques adjust the model's weights and biases iteratively to find the optimal set of parameters that reduces the error between predicted and actual values (Alzubaidi et al., 2021).

Different optimizers are used to update model weights efficiently:

- **Stochastic Gradient Descent (SGD):** Updates weights based on a small subset (batch) of training data, improving computational efficiency.
- **Adam (Adaptive Moment Estimation):** Combines momentum and adaptive learning rates for faster and more stable convergence.
- **RMSprop:** Uses an adaptive learning rate to prevent oscillations and improve performance on non-stationary objectives.

2.2.5 Model evaluation and validation

Once a model is trained, it is crucial to assess its performance and ensure that it generalizes well to unseen data. This process is known as **model evaluation and validation**. The primary goal is to measure how well the model performs and to identify any potential overfitting or underfitting issues.

To evaluate a model's performance, several metrics can be used depending on the type of task (classification, regression, etc.) (Ketkar & Santana, 2017):

- **Accuracy:** For classification tasks, accuracy measures the proportion of correct predictions out of all predictions made. However, accuracy alone can be misleading in imbalanced datasets.
- **Precision and Recall:** In imbalanced classification problems, precision (the proportion of true positive results among all positive predictions) and recall (the proportion of true positive results among all actual positives) are often used in conjunction to provide a clearer view of the model's performance.
- **F1-Score:** The harmonic mean of precision and recall; F1-score balances the two metrics and is especially useful when dealing with imbalanced datasets.
- **Mean Squared Error (MSE):** For regression tasks, MSE calculates the average of the squared differences between predicted and actual values. It penalizes large errors more significantly than smaller ones.
- **R-squared (R^2):** For regression, this metric indicates how well the model explains the variability in the data, with a value closer to 1 suggesting a better fit.

2.3 Core computer vision tasks in agriculture

The most common and impactful computer vision tasks include image classification, image segmentation, and object detection. Each of these tasks plays a vital role in applications such as disease diagnosis, crop monitoring, yield estimation, and weed or pest detection. The following sections explore these tasks and their corresponding deep learning techniques in detail.

2.3.1 Image classification

Image classification is the task of assigning a label to an entire image based on its visual content. In deep learning, this is typically achieved using CNNs, which are designed to automatically learn hierarchical features from raw pixel data.

Fundamentals of CNNs

CNNs (Figure 2.3) consist of multiple layers designed to process and learn spatial hierarchies from image data (Alzubaidi et al., 2021). Their architecture typically includes:

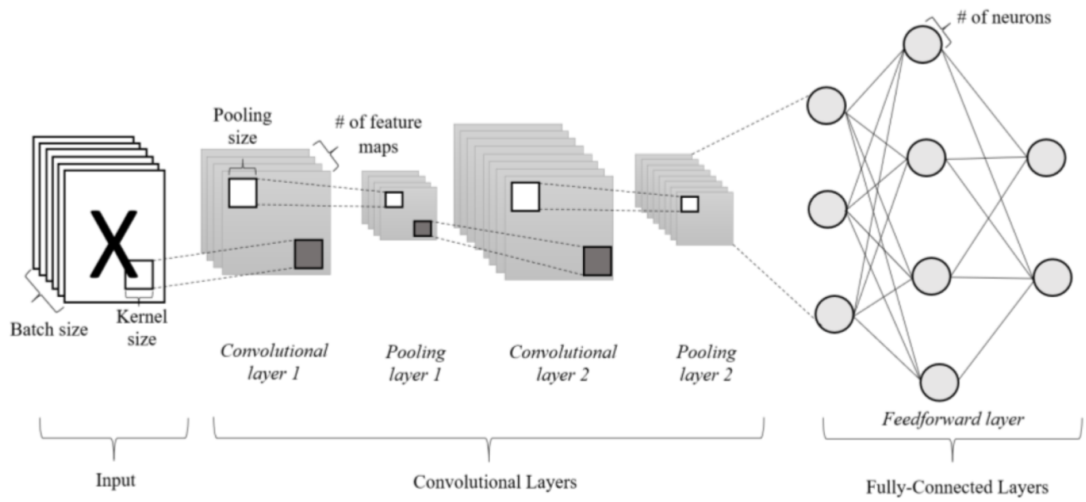

Figure 2.3: Architecture of CNN (Ang et al., 2022).

- **Convolutional Layers:** Extract features using small filters (kernels) that detect edges, textures, and patterns. A kernel is a grid of values (weights) initialized randomly at the start of training and adjusted through learning to identify important features.
  - **Input and Kernel Dimensions:** In a CNN layer, each input x is structured in three dimensions: height, width, and depth. The depth corresponds to the number of channels (e.g., an RGB image has three channels). Similarly, the kernels are also three-dimensional, with spatial dimensions (height and width) and a depth matching the input channels. Each kernel has shared parameters, a set of weights and a bias. When applied to the input, these kernels generate a corresponding set of feature maps. These kernels establish local connections by interacting only with small regions of the input at a time, allowing the network to extract patterns such as edges and textures by computing dot products across these regions.
  - **Convolutional operation:** Mathematically, a 2D convolution operation between an input matrix I and a kernel (or filter) K is expressed as shown in equation 2.5:

$$S(i,j) = \sum_m \sum_n I(i+m, j+n) \cdot K(m,n) \tag{2.5}$$

    where $S(i,j)$ is the output feature map value at position (i,j), I is the input image, and K is the kernel.

    The convolutional process involves sliding the kernel across the input image in both horizontal and vertical directions. At each location, the dot product between the kernel and the overlapping region of the input is computed, producing a scalar value that becomes part of the resulting feature map. As this process is repeated across the image, a full feature map is constructed, highlighting areas where the kernel detects specific patterns. Parameters such as stride (which controls how far the kernel moves at each step) and padding (which adds borders to the input to preserve edge information) affect the size and coverage of the output feature map. These concepts are illustrated in Figure 2.4.

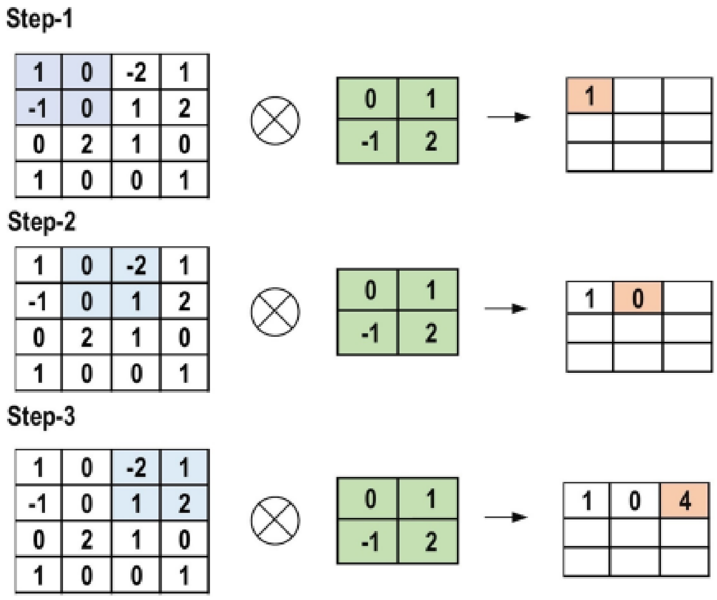

Figure 2.4: The primary calculations executed at each step of the convolutional layer (Alzubaidi et al., 2021).

- **Pooling Layers:** Pooling layers are used to reduce the spatial dimensions of feature maps while preserving the most important information. This reduction helps lower the computational cost and minimizes the risk of overfitting by simplifying the data representation. The pooling operation works by sliding a small filter over the feature map and applying a summary function within each local region (Figure 2.5). Common types of pooling include:
  - **Max Pooling:** Selects the maximum value in each region.
  - **Average Pooling:** Calculates the average value of the region.
  - **Global Average Pooling:** Computes the average across the entire feature map.

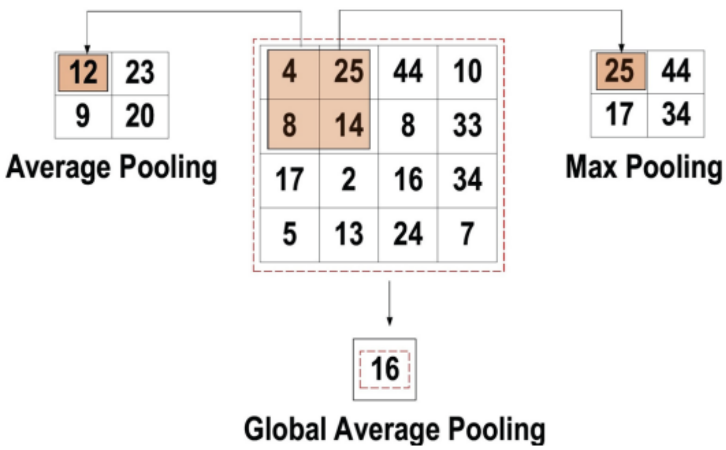

Figure 2.5: Three types of pooling operations (Alzubaidi et al., 2021).

- **Fully Connected Layers:** The Fully Connected (FC) layer is typically found at the end of a CNN architecture and serves as the classifier. In this layer, each neuron is connected to all neurons from the previous layer, following the fully connected approach. It operates similarly to a conventional multi-layer perceptron (MLP) network, which is a type of feed-forward artificial neural network (ANN). The input to the FC layer is a vector created from the feature maps after flattening, which comes from the last pooling or convolutional layer. The output of the FC layer represents the final result of the classification task.

Convolutional Neural Network Architectures for Image Classification

Over the past decade, various Convolutional Neural Network (CNN) architectures have been developed, each incorporating unique design principles to enhance accuracy, efficiency, and scalability. In image classification tasks particularly in plant disease detection the choice of architecture plays a crucial role, influenced by dataset size, complexity, and computational constraints.

Among these, lightweight and specialized models such as MobileNet, EfficientNet, and ShuffleNet have been designed to address speed, size, and energy limitations, making them ideal for mobile and edge device deployment. A comparative overview of both standard and lightweight CNN architectures, along with their key characteristics and potential use in smart agriculture, is provided in Appendix 3.9.

Transfer learning and pretrained models

Transfer learning is a machine learning technique where a model trained on one task is repurposed for a different but related task. In the context of deep learning, it typically involves taking a neural network pre-trained on a large dataset such as ImageNet, which contains over 14 million labeled images, and adapting it to a specific task that may lack sufficient labeled data.

To formalize this, consider a target learning task T_t based on a domain D_t; transfer learning allows for assistance from a different domain D_s for the learning task T_s. The goal of transfer learning is to improve the performance of the predictive function $f_{T_t}(\cdot)$ for the task T_t by discovering and transferring latent knowledge from D_s and T_s, where generally $D_s = D_t$ and/or $T_s = T_t$. Furthermore, it is often the case that the size of D_s is much larger than that of D_t (Tan et al., 2018).

The two primary approaches to transfer learning are (Tan et al., 2018):

- **Feature Extraction:** The pre-trained model is used as a fixed feature extractor. All convolutional layers are kept frozen, and only the final fully connected layer(s) are trained on the new dataset.
- **Fine-Tuning:** Some layers of the pretrained model are unfrozen and retrained on the new dataset. This allows the model to slightly adjust its learned features to better suit the new domain.

2.3.2 Object detection

In the context of precision agriculture, object detection has emerged as a critical computer vision technique for automating the assessment of crop health by enabling the identification and localization of diseases, nutrient deficiencies, and weeds, thereby supporting more efficient and timely interventions on a large scale.

At its core, object detection involves predicting both the category and precise location of objects within an image, thus combining classification with localization. Traditional approaches consist of stages such as region proposal, feature extraction, and object classification. Over time, detection methods have evolved to address increasing demands for accuracy and speed, giving rise to both two-stage and one-stage detection frameworks (Zhao et al., 2019).

Here are more details about the concepts of object detection (Zhao et al., 2019):

- **Informative Region Selection** focuses on identifying likely object areas in an image to reduce computation by avoiding full image scans. While early methods like sliding windows were costly and redundant, modern approaches use region proposal algorithms or attention mechanisms to efficiently target relevant areas, especially useful in agricultural images with variable object sizes and shapes.

- **Feature Extraction** is the process of isolating important visual attributes from an image to effectively represent objects. Traditional methods, such as Scale Invariant Feature Transform (SIFT), Histograms of Oriented Gradients (HOG), and Haar-like features were designed to mimic human vision by focusing on edges, textures, and patterns. However, these handcrafted techniques often faced difficulties in real-world conditions due to variations in object appearance, lighting, and cluttered backgrounds, limiting their reliability in complex environments.

- **Classification and Localization** refer to the process of both identifying the object in an image and determining its precise location through bounding boxes. With the rise of deep learning, this step underwent a significant transformation. Models such as R-CNN and its subsequent versions: Fast R-CNN, Faster R-CNN, and YOLO automated the feature extraction process and seamlessly integrated classification with bounding box regression. These advancements led to substantial improvements in both detection accuracy and processing speed, enabling real-time applications in fields like crop monitoring and pest detection.

Evaluation metrics in object detection

In object detection, several metrics are used to assess model performance:

- **Mean Average Precision (mAP):** Measures the average precision across all object classes, balancing precision and recall to evaluate overall model performance.

- **Intersection over Union (IoU):** Calculates the overlap between predicted and ground truth bounding boxes, indicating localization accuracy. Higher IoU means better localization.

- **Precision:** Measures the proportion of true positive detections out of all predicted objects.

- **Recall:** Measures the proportion of true positive detections out of all actual objects.

- **F1-Score:** The harmonic mean of precision and recall, providing a balance between both.

- **Average Recall (AR):** Evaluates recall at different IoU thresholds, useful for detecting small objects or handling occlusions.
- **Confusion Matrix:** Summarizes true positives, false positives, true negatives, and false negatives, providing insights into model errors.
- **Speed Metrics** like frames per second (FPS) and latency measure how fast an object detection model works in real time. FPS shows how many images the system can handle each second, higher FPS means smoother, faster image analysis. Latency is the time it takes to process one image and give a result, lower latency means quicker response.
- **AP at Specific IoU Thresholds:** Measures precision at different IoU levels to understand performance under stricter conditions.

Key architectures

Many architectures are designed to efficiently and accurately detect objects in images, even in complex agricultural environments. Below, we discuss four of the most widely used and effective object detection models: YOLO, R-CNN, and SSD.

You Only Look Once (YOLO)

YOLO is a fast and efficient object detection framework that predicts both object confidences and bounding boxes (BBs) using the entire topmost feature map. The image is divided into a $S \times S$ grid, where each grid cell is responsible for predicting objects centered within it. Each cell predicts multiple bounding boxes and their corresponding confidence scores, which reflect the likelihood of an object being present and how well the predicted box overlaps with the ground truth (IoU) (see figure 2.6).

At test time, class-specific confidence scores are computed by multiplying the box confidence with conditional class probabilities. YOLO optimizes a loss function during training to fine-tune predictions and improve detection accuracy (Zhao et al., 2019).

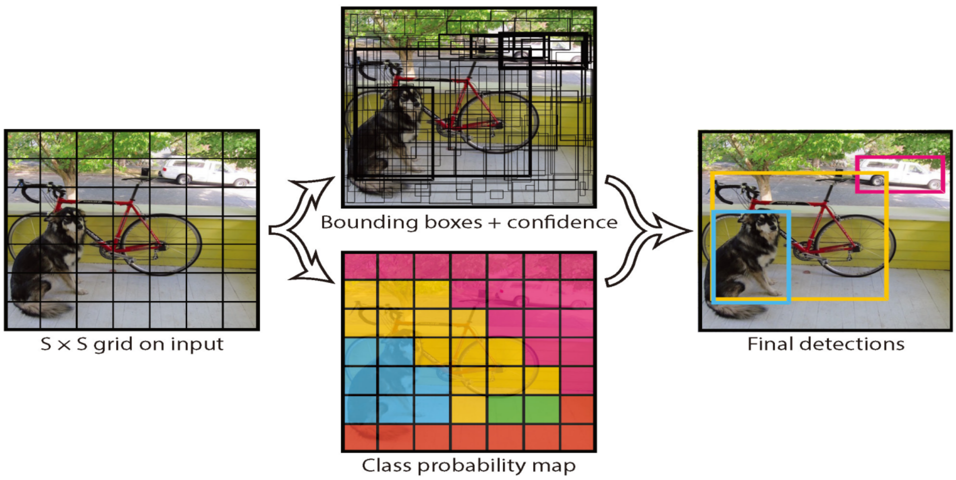

Figure 2.6: Main idea of YOLO (Zhao et al., 2019).

R-CNN

R-CNN is a significant advancement in object detection, improving the quality of candidate bounding boxes (BBs) and utilizing deep architecture for high-level feature extraction (Zhao et al., 2019). It consists of three main stages, as presented in figure 2.7:

- **Region Proposal Generation:** R-CNN uses selective search to generate about 2000 region proposals per image, improving candidate box accuracy and reducing the search space.
- **CNN-Based Feature Extraction:** Each region proposal is resized and passed through a CNN to extract a 4096-dimensional feature, creating a high-level, robust representation of the object.
- **Classification and Localization:** Region proposals are classified using pre-trained linear SVMs, and bounding box regression is applied. Non-maximum suppression (NMS) is used to eliminate redundant boxes and finalize object detections.

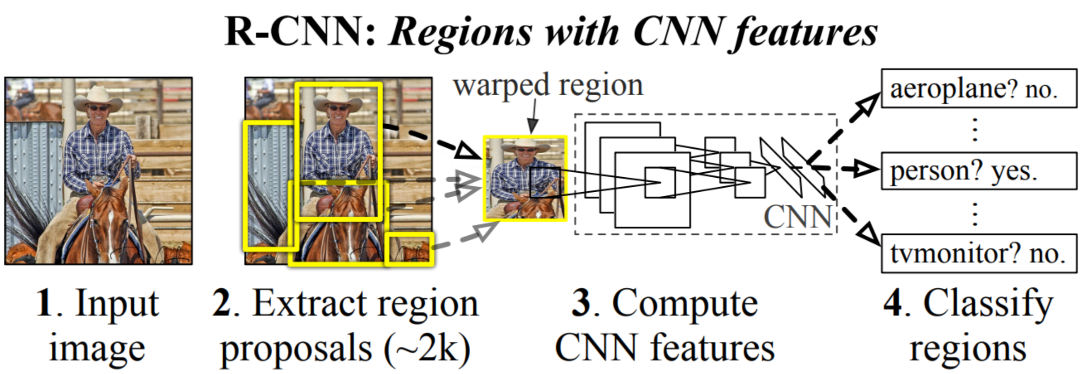

Figure 2.7: Flowchart of R-CNN (Zhao et al., 2019).

Despite its success, R-CNN has drawbacks, including slow inference due to CNN computation for each region, time-consuming multistage training, high memory and storage requirements for storing region features, and redundant region proposals from selective search that slow down the process (Zhao et al., 2019).

Single Shot Multibox Detector (SSD)

The SSD model was developed to overcome limitations in YOLO, especially with small and irregularly shaped objects. Unlike YOLO's fixed grid strategy, SSD utilizes multiple default anchor boxes of varying aspect ratios and scales, enabling better detection across object sizes. It combines predictions from multiple resolution feature maps and employs a VGG16 backbone with added layers for object localization and classification (see Appendix 3.9). SSD uses a combination of localization and confidence loss during training and refines results using non-maximum suppression. It achieves higher accuracy than Faster R-CNN on PASCAL VOC and COCO, while being significantly faster at 59 fps for 300×300 input, though performance on small objects remains an area for improvement (Zhao et al., 2019).

The architecture of SSD is illustrated in Figure 2.8, highlighting the VGG16 backbone, additional feature layers, and classifier convolutions.

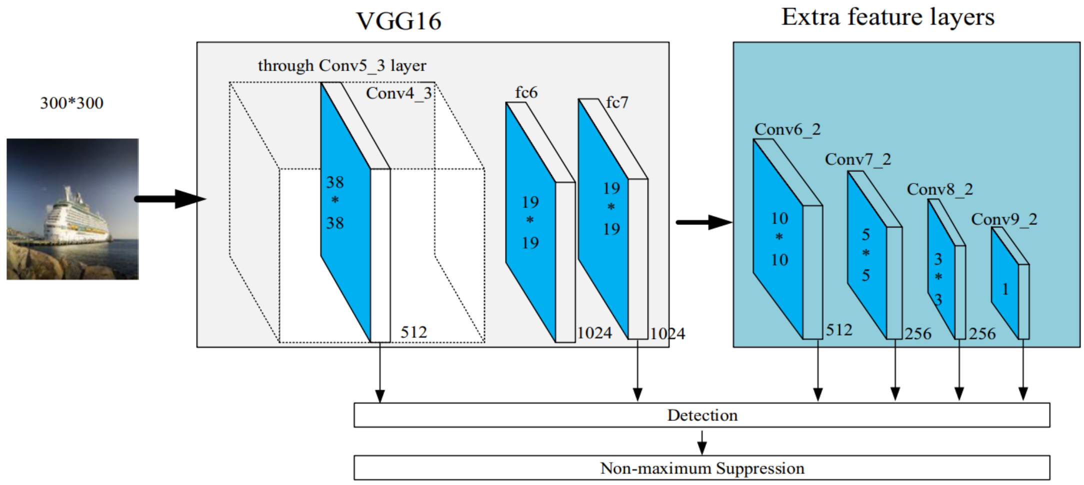

Figure 2.8: Architecture of SSD (A. Li et al., 2021).

2.3.3 Image segmentation

Image segmentation is a computer vision task that involves partitioning an image into multiple segments, or regions, to make the analysis of its content more manageable. Each pixel is classified into a specific category, which helps in identifying objects or boundaries within an image. In wheat disease detection, image segmentation is crucial for isolating affected areas and distinguishing between healthy and diseased plant tissues. Models like PSPNet, U-Net and its variants, and DeepLabV3+ are particularly effective for this task, as they can capture both fine details and global context, leading to more accurate disease detection.

Pyramid Scene Parsing Network (PSPNet)

PSPNet is a deep convolutional neural network architecture designed for semantic segmentation tasks, particularly effective in capturing both local and global contextual information. The core innovation of PSPNet lies in its Pyramid Pooling Module (PPM), which aggregates context from different regions of the image at multiple scales. This is achieved by applying pooling operations at varying grid sizes, then upsampling and concatenating these pooled features with the original feature maps (Pan et al., 2021). This fusion allows PSPNet to understand the spatial hierarchy and global structure of scenes more effectively than traditional CNNs (Figure 2.9). residual network (ResNet) and its variants are usually used as the backbone for PSPNet due to their residual (skip) connections, which help alleviate the vanishing gradient problem and enable effective training of deep networks, making them well-suited for extracting rich feature representations in segmentation tasks.

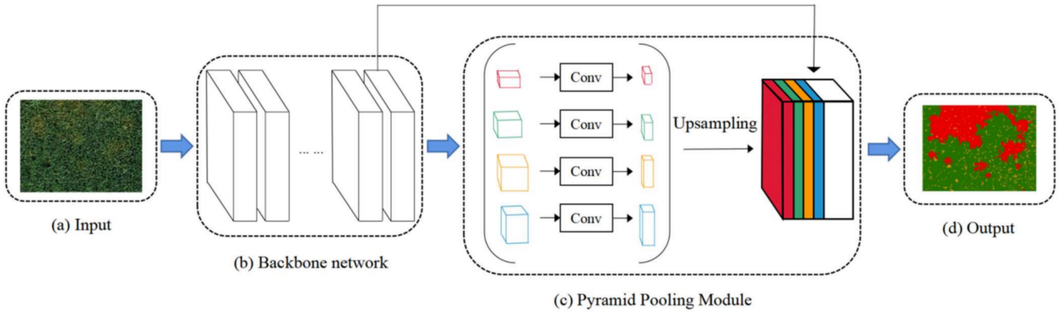

Figure 2.9: Semantic Segmentation of Wheat Yellow Rust Using PSPNet (Pan et al., 2021).

U-Net model architecture

U-Net is a convolutional neural network architecture designed for semantic segmentation, featuring a symmetric encoder-decoder structure. The encoder, or contracting path, captures contextual features using repeated 3×3 convolutions with ReLU activation, followed by 2×2 max pooling for downsampling. The decoder, or expansive path, performs upsampling and applies 3×3 convolutions to recover spatial resolution. Crucially, skip connections between corresponding encoder and decoder layers preserve high-resolution features, improving segmentation accuracy. A final 1×1 convolution maps the feature maps to the desired number of output classes. All convolutions use "same" padding with stride [1,1], and max pooling uses stride [2,2] with zero padding. ReLU activation is applied throughout the network, and input images undergo zero-center normalization for training stability (Su et al., 2020). This architectural layout, including feature map sizes and layer depths, is depicted in Figure 2.10.

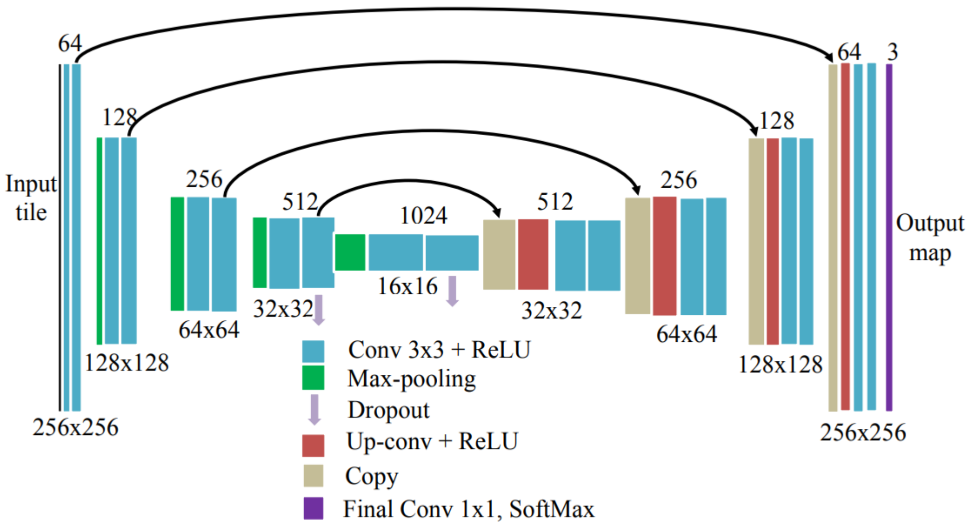

Figure 2.10: U-Net Architecture for Semantic Segmentation (Su et al., 2020).

Several U-Net variants have been proposed to address computational efficiency and feature enhancement:

- **LR-UNet (Lightweight Residual U-Net)**: This architecture introduces residual connections and lightweight convolutional blocks to reduce the number of parameters and computational cost while maintaining segmentation performance. It is especially suitable for deployment in resource-constrained environments such as mobile or embedded devices (T. Zhang et al., 2021).

- **DF-UNet (Dual Feature U-Net)**: This model enhances the original U-Net by fusing both shallow (low-level spatial) and deep (high-level semantic) features. This dual feature strategy improves the network's ability to detect fine edges and preserve structural details, particularly in complex segmentation tasks such as medical or remote sensing imagery. DF-UNet may also integrate attention mechanisms and refined skip connections for more effective feature reuse (T. Zhang et al., 2022).

DeepLabV3+

DeepLabV3+ is an advanced semantic segmentation model that builds upon DeepLabV3 by combining powerful contextual feature extraction with precise boundary recovery. DeepLabV3 uses Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual information through parallel atrous (dilated) convolutions with varying rates, allowing the model to extract rich semantic features without reducing spatial resolution too much. However, DeepLabV3 lacks a decoder and relies on simple bilinear upsampling, which limits its ability to recover fine object boundaries. DeepLabV3+ (Figure 2.11) addresses this by introducing a decoder module that refines segmentation results, especially along object edges, by upsampling the encoder's output and combining it with low-level features from earlier layers of the network, followed by a few convolutions for detail refinement. Additionally, DeepLabV3+ applies depthwise separable (atrous separable) convolutions in both the ASPP and decoder modules to reduce computational cost while maintaining high accuracy, resulting in a more accurate and efficient architecture (Chen et al., 2018).

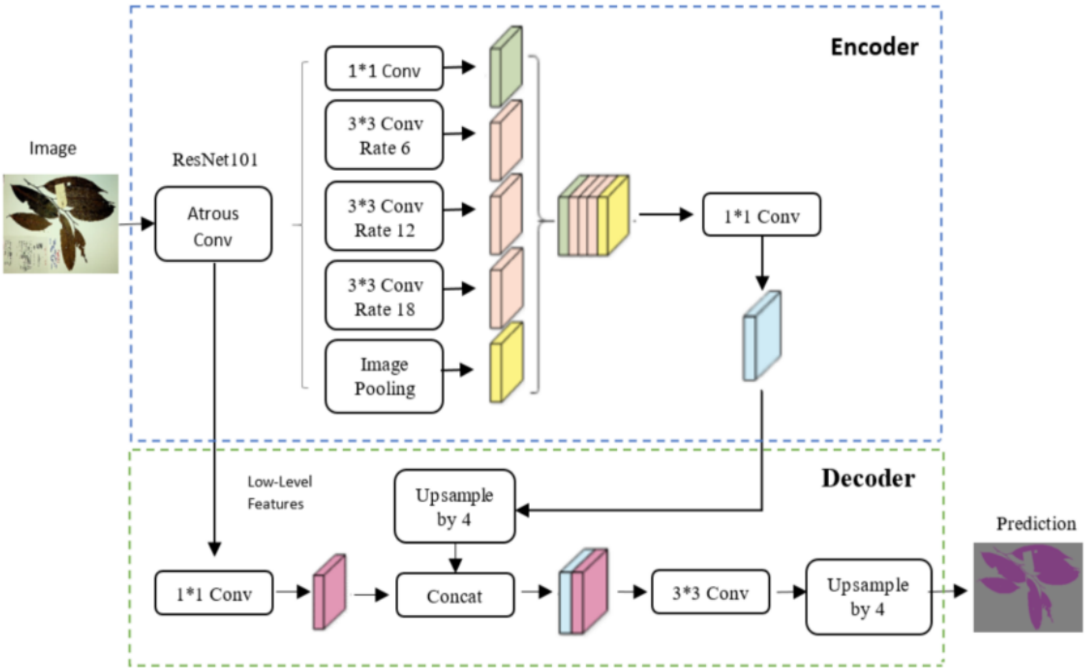

Figure 2.11: DeepLabv3+ architecture (Hussein et al., 2020).

2.4 Conclusion

In this chapter, we explored key deep learning concepts and models used for image-based recognition tasks, focusing on classification, object detection, and semantic segmentation. These techniques are essential for automating and improving agricultural workflows, especially in crop health monitoring. Building on this foundation, The next chapter explores how these techniques can be integrated with remote sensing data for early and accurate detection of wheat diseases.

Chapter 3

Integrating Remote Sensing and AI-Based Methods for Early Detection of Wheat Diseases

3.1 Introduction

Wheat is one of the most essential cereal crops globally, and its productivity is highly threatened by various diseases. Traditional methods of crop monitoring often fall short in providing timely and accurate detection, especially at scale. The integration of remote sensing technologies with artificial intelligence (AI) approaches such as machine learning (ML) and deep learning (DL) opens new possibilities for precision agriculture.

This chapter aims to explore how these complementary tools can be combined to monitor wheat health and detect diseases early. We begin by outlining the motivation behind this integration, reviewing the major types of wheat diseases, and then discussing the methodologies and workflows for combining remote sensing data with intelligent AI-based analysis.

3.2 Motivation for the protection of wheat crops

Wheat is an ancient and vital food crop that provides energy and feeds billions of people around the world (see Figure 3.1). Its demand is growing quickly because it's used in many affordable food products and plays a big role in global food security. The food and agriculture organization (FAO) estimates that by 2050, the world will need about 840 million tonnes of wheat, up from 642 million tonnes today (Sharma et al., 2015). This doesn't even include the extra needs like animal feed or the impact of climate change.

The global importance of wheat is further reflected in its massive export volumes, with leading producers shipping tens of millions of tonnes annually (see Table 3.1), demonstrating its critical role in both domestic consumption and international food supply chains. To meet the growing demand, developing countries need to increase wheat production by 77%, mostly by improving how much wheat is grown on the same land (Sharma et al., 2015). But this is becoming harder, as wheat productivity is slowing down and diseases are becoming a bigger problem. If we don't manage pests and diseases properly, wheat production could fall short of what the world needs.

That's why it's important to invest in research, use better farming methods, and grow disease-resistant wheat. Protecting this essential crop is key to making sure we have enough food for the future.

Table 3.1: Top 10 Countries Exporting the Most Wheat (2022) (World Population Review, 2024)

Country	**Wheat Exports 2022 (T)**
Australia	28.8M
United States	20.9M
France	20.2M
Canada	18.5M
Russia	17.8M
Argentina	12.9M
Ukraine	11.2M
India	6.8M
Kazakhstan	6.3M
Germany	6.2M

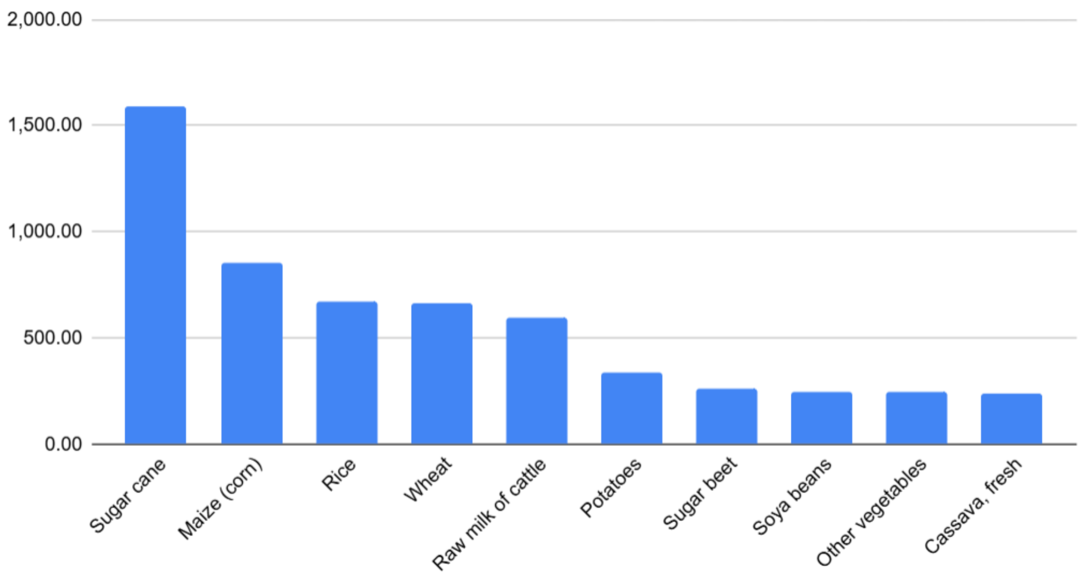

Figure 3.1: Most Produced Commodities Worldwide in 2023 (in Million Tonnes) (FAO, 2025)

The significance of wheat is also evident in its production across Arabic speaking nations, As shown in the figure 3.2 below, with Algeria ranking as the second-largest producer at 7 million tonnes per year, just behind Egypt (9.5 million tonnes). While this output is significant, production could increase even further by adopting modern farming techniques like the integrating of smart agriculture solutions.

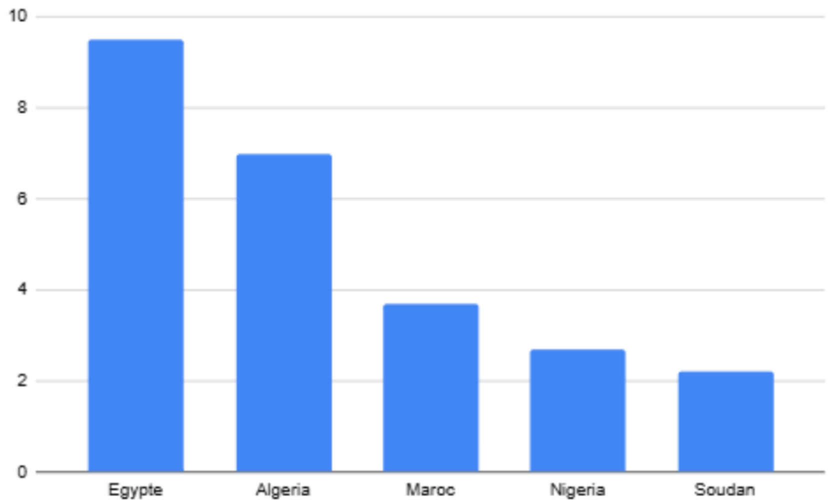

Figure 3.2: Main African wheat producers in 2022/2023 (Tadamoun News, 2024).

3.3 Wheat growth stages

Wheat growth follows a series of distinct developmental stages (Figure **??**), each with specific environmental needs and a direct impact on crop health and yield (Shafi et al., 2020). The key stages are:

- **Seeding:** The initial stage where seeds germinate and seedlings emerge. Adequate soil moisture and suitable temperatures are essential for successful crop establishment.
- **Tillering:** The plant produces side shoots (tillers), which increase potential yield. This stage depends on sufficient nutrients and water.
- **Booting:** The wheat head develops inside the leaf sheath. Stress at this stage can reduce the number of spikelets and affect grain development.
- **Heading:** The spike emerges and flowering occurs. This is a sensitive period where drought or heat can severely impact pollination and grain set.
- **Ripening:** Grains mature and the plant loses its green color. Proper conditions are needed for effective grain filling and harvest readiness.

3.4 Wheat diseases: types and impacts

Wheat is affected by various diseases that can reduce yield and grain quality. These diseases, mainly caused by fungi, thrive under specific environmental conditions and target different parts of the plant. This section outlines the major types of wheat diseases, their symptoms, causes, and impacts on crop production.

3.4.1 Leaf rust (Brown Rust)

Leaf rust, caused by *Puccinia triticina*, appears as small, circular, orange to brown pustules on the upper surfaces of leaves and leaf sheaths. It spreads through wind-borne spores and develops quickly in moist conditions at around 20°C. New spores form every 10–14 days if conditions are favorable. As plants mature or conditions worsen, black spores may appear (as shown in Figure 3.3).
This disease affects wheat, triticale, and related grasses and is common in temperate cereal-growing regions. Severe infections reduce grain yield, kernel number, weight, and quality (Duveiller et al., 2012).

Figure 3.3: Leaf rust with Brown spores (1), Leaf rust with Black spores (2) (Duveiller et al., 2012)

3.4.2 Stem rust (Black Rust)

Stem rust, caused by *Puccinia graminis*, appears as dark reddish-brown pustules on leaves, stems, and spikes (as observed in Figure 3.4). Light infections show scattered pustules, while severe cases cause them to merge. Before pustules form, small flecks may appear, and infected areas feel rough. The disease spreads through wind-borne spores and develops quickly in moist conditions with temperatures around 20°C. New spores can form in 10–15 days. It affects wheat, barley, triticale, and related grasses and is common in temperate cereal regions. Severe infections can reduce grain weight and quality and, in extreme cases, lead to total crop loss (Duveiller et al., 2012).

Figure 3.4: Stem rust (Duveiller et al., 2012)

3.4.3 Stripe rust (Yellow Rust)

Stripe rust, caused by *Puccinia striiformis*, appears as yellow to orange-yellow pustules forming narrow stripes on leaves, leaf sheaths, necks, and glumes (as seen in Figure 3.5). It spreads through wind-borne spores and develops quickly in moist conditions at temperatures between 10–20°C but slows down above 25°C. Severe infections reduce grain yield, kernel number, weight, and quality (Duveiller et al., 2012).

Figure 3.5: Stripe rust (Duveiller et al., 2012)

3.4.4 Blotch diseases

The blotch diseases, which include Septoria tritici blotch (STB), Septoria nodorum blotch (SNB), and tan spot (TS) (as presented in Figure 3.6), are caused by the *Ascomycete fungi Zymoseptoria tritici*, *Parastagonospora nodorum*, and *Pyrenophora tritici-repentis*, respectively (Figueroa et al., 2018).

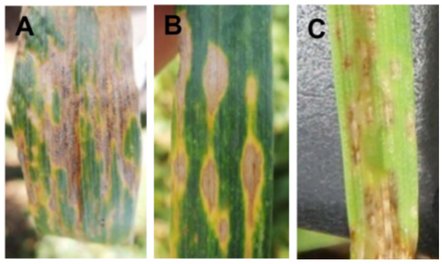

Figure 3.6: Symptoms of foliar blotch diseases. (A) Septoria tritici blotch. (B) Tan spot. (C) Septoria nodorum blotch (Figueroa et al., 2018)

3.4.5 Fusarium head blight (FHB)

Fusarium head blight (FHB), also known as wheat scab or ear blight, is a major disease of wheat caused primarily by the Ascomycete fungus *Fusarium graminearum* (Fg). It can also be caused by other regional *Fusarium* species (Figueroa et al., 2018).
Fusarium head blight appears as dark, oily florets with pinkish spores (as seen in Figure 3.7). Infected kernels may be covered in white fungal growth. The disease spreads in warm, humid conditions (10–28°C), infecting spikes during flowering and spreading between florets. It affects all small grain cereals and is found in most soils and crop residues. Severe infections can reduce yields by over 50% and lower grain quality. Contaminated grain may contain harmful mycotoxins, making it unsafe for humans and animals (Duveiller et al., 2012).

Figure 3.7: Symptoms of Fusarium head blight/scab: The left image shows early infection signs manifested as a partially bleached wheat head, while the right image illustrates advanced infection of *Fusarium graminearum* (Figueroa et al., 2018).

3.4.6 Loose smut

Loose smut, caused by *Ustilago tritici*, replaces wheat spikes with black fungal spores (as observed in Figure 3.8), which are later dispersed by wind. The fungus infects wheat flowers and stays dormant in kernels until germination. It then grows with the plant, destroying floral parts at flowering. The disease thrives in cool, humid conditions and is found wherever wheat is grown. Yield losses depend on infection levels, usually below 1% but sometimes reaching 30% (Duveiller et al., 2012).

Figure 3.8: Loose smut (Duveiller et al., 2012)

3.4.7 Powdery mildew

Powdery mildew (Figure 3.9), caused by *Blumeria graminis f. sp. tritici*, affects wheat globally, particularly in cool, dry climates, and can cause yield losses ranging from 10% to 40%, with severe cases leading to seedling or tiller death (Singh et al., 2023).

Figure 3.9: Powdery mildew from Kaggle dataset 'Wheat Plant Diseases.'

3.4.8 Common root rot

Common root rot, caused by *Cochliobolus sativus*, *Fusarium spp*, and *Pythium spp*, darkens and weakens wheat roots, crowns, and stems (as illustrated by Figure 3.10), sometimes leading to plant lodging and white spikes before maturity. Early infections can cause seedling death. The disease spreads from infected crop debris, thriving in different soil conditions: *C. sativus* in warm, dry soils and *Fusarium* and *Pythium* in cool, moist soils. Found in temperate regions, it rarely causes major outbreaks but can lead to localized losses due to reduced plant growth and yield (Duveiller et al., 2012).

Figure 3.10: Common root rot from Kaggle dataset 'Wheat Plant Diseases.'

3.5 Towards Scalable and Reliable Crop Disease Detection via AI and Remote Sensing

While deep learning has demonstrated remarkable performance across numerous domains, its application in agriculture particularly in crop disease detection presents distinct challenges. Key limitations include the scarcity and imbalance of annotated datasets, high intra-class variability, and low inter-class variability, all of which hinder robust model generalization. Additionally, real-world agricultural images are often affected by variations in lighting, occlusion from overlapping leaves, and cluttered backgrounds, further reducing detection accuracy (Alzubaidi et al., 2021).

A significant constraint is the reliance of most models on close-range imagery, which limits their effectiveness in early-stage detection across large-scale farmlands. To overcome these issues, recent research has increasingly focused on integrating remote sensing technologies with AI-based models. Remote sensing using data from satellites, drones (UAVs), and hyperspectral or multispectral sensors, provides valuable spectral, thermal, and spatial information across broad geographic areas. When combined with deep learning, this fusion enables more accurate early detection, classification, and spatial mapping of plant diseases, even before visible symptoms emerge. Such integration not only enhances detection accuracy but also supports scalable, timely, and cost-effective crop health monitoring and management strategies (Wang et al., 2024).

3.6 Workflow for integrating remote sensing and ML/DL in wheat disease detection

Existing research on crop disease detection using UAV imagery can be grouped into three main methodological approaches (Figure 3.11). The first involves statistics-based methods, which apply correlation and regression analyses to identify linear relationships between disease symptoms and spectral data from UAV images. These methods commonly use vegetation indices (vegetation index (VI)s) to extract relevant crop traits. The second approach includes conventional machine learning techniques, where VIs serve as input features for supervised or unsupervised models used in disease estimation. The third and most recent approach is based on deep learning, which leverages raw UAV images alongside other features to train end-to-end models for automated crop disease recognition (Shahi et al., 2023).

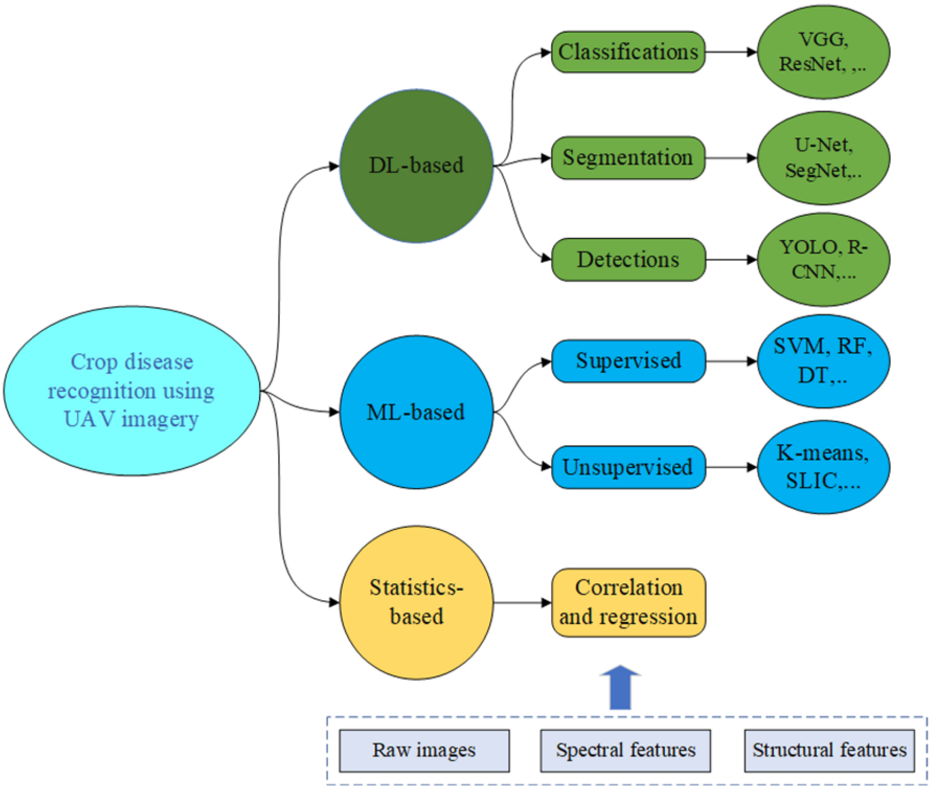

Figure 3.11: Classification of crop disease detection approaches using UAV-based remote sensing. The elements within the dotted box represent the image features utilized across one or more of the listed approaches (Shahi et al., 2023).

3.6.1 Features extraction

In the context of remote sensing for plant disease detection, different types of features can be extracted from imagery to capture various physiological, structural, and visual properties of crops.

- **Spectral Features**: Describe how plants reflect light across different spectral bands (such as visible, red edge, or near-infrared). These features reveal the biochemical state of vegetation, helping detect conditions like chlorophyll content, moisture level, and early stress symptoms using vegetation indices (e.g., NDVI, NDRE) or raw spectral band reflectance (Dhakal et al., 2023).
- **Structural Features**: Represent the physical shape, size, and arrangement of plant components, such as canopy height, leaf area, or plant density. They help monitor growth patterns and identify visible deformations caused by disease, pests, or environmental stress through 3D models or spatial measurements (Dhakal et al., 2023).
- **Textural Features (TF)**: Capture the variation and arrangement of pixel intensities in an image, describing surface patterns like roughness, smoothness, or regularity. These features are useful for detecting visual symptoms of disease, such as lesions or uneven leaf surfaces, using statistical metrics like contrast, entropy, and homogeneity (Dhakal et al., 2023).

- **Wavelet Features (WF)**: Capture spatial and spectral variations in data by breaking it down into components at different scales. These features can detect fine-scale patterns, which are useful for identifying subtle changes in vegetation, such as variations caused by disease or stress (Ma et al., 2021).

3.6.2 Statistics-Based methods

Statistics-based methods for crop disease estimation rely on using crop-related traits derived from UAV imagery as independent variables and disease scores as the target variable. These methods typically involve three main steps: image pre-processing, vegetation index (VI) generation, and statistical analysis. During pre-processing, spatial data products such as reflectance maps are generated, representing how much light vegetation reflects across different wavelengths, which is crucial for assessing plant health. These maps calculate various vegetation indices (e.g., NDRE, NDVI, and DVI) to summarize plant conditions like chlorophyll content, stress, or canopy vigor. These indices are then used in correlation and regression analyses to estimate disease severity (Shahi et al., 2023).

In practice, fields are often divided into small plots, and the mean VI value per plot is computed for analysis. Some studies also apply threshold-based techniques, setting a cutoff value to distinguish between healthy and diseased areas. However, defining a universal threshold is difficult due to variability in crop type, disease symptoms, and imaging conditions.

For example, (Guo et al., 2021) employed UAV-based hyperspectral imagery to detect wheat yellow rust and incorporated both VIs and texture feature (TF)s into their statistical models. (Guo et al., 2021) extracted TFs using Principal Component Analysis (principal component analysis (PCA)), followed by methods like the Gray-Level Co-occurrence Matrix (gray-level co-occurrence matrix (GLCM)), a technique used to analyze the spatial relationship between pixel intensities in an image. GLCM captures texture features such as contrast, entropy, and correlation, which help detect subtle patterns in plant surfaces that could indicate disease or stress. By combining these TFs with spectral indices in a Partial Least Squares Regression (partial least squares regression (PLSR)) model, they achieved significantly higher accuracy, reaching an R^2 of 0.88 in the late infection stage. PLSR is a regression technique that is particularly effective when the predictors are many and possibly collinear, as it reduces dimensionality and finds the most relevant directions for predicting the response variable. The general form of the PLSR equation (3.1) is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n \tag{3.1}$$

where Y is the disease severity index, $X_1, \ldots, X_n$ are the input features (e.g., VIs, TFs), and $\beta_0, \ldots, \beta_n$ are the regression coefficients determined during model training.

Their study also highlighted that high spatial resolution (10–15 cm) images were critical for capturing fine structural differences caused by disease. Similarly, (Bhandari et al., 2020) achieved a strong correlation ($R^2 = 0.79$) using the GLI index derived from RGB images to assess foliar diseases in wheat (Table 03). Together, these studies show that combining spectral and textural information enhances the performance of statistical models in UAV-based crop disease monitoring.

The detailed formulas for the vegetation indices (VIs) used in the studies mentioned are provided in Appendix 3.9.

Table 3.2: Summary of ST-based methods for Wheat disease estimation using UAV imagery.

Paper	Disease	Sensor	Vegetation Indices (VIs)	Eval. Metrics
(Bhandari et al., 2020)	Wheat foliage disease	RGB	NDI, GI, GLI	$R^2 = 0.79$
(Heidarian Dehkordi et al., 2020)	Leaf rust, Stem rust	RGB	SRI, LRI	$R^2 = 0.81$
(Guo et al., 2021)	Yellow rust	Hyperspectral	SIPI, PRI, TCARI, PSRI, YRIGI	$R^2 = 0.88$
(Khan et al., 2021)	Powdery Mildew	Hyperspectral	PMI, MSR, MCARI	$R^2 = 0.722$

3.6.3 Conventional machine learning (ML)-Based methods

Traditional machine learning (ML) methods, including support vector machines (support vector machine (SVM)s), artificial neural networks (ANNs), Support Vector Regression (support vector regression (SVR)), have been applied to crop disease detection using UAV imagery. These methods aim to identify patterns in labeled or unlabeled data, with supervised learning relying on labeled datasets for training and unsupervised learning exploring hidden patterns without labels (Shahi et al., 2023). The typical ML pipeline involves data collection, preprocessing, feature extraction, and model building, as shown in Figure 3.12.

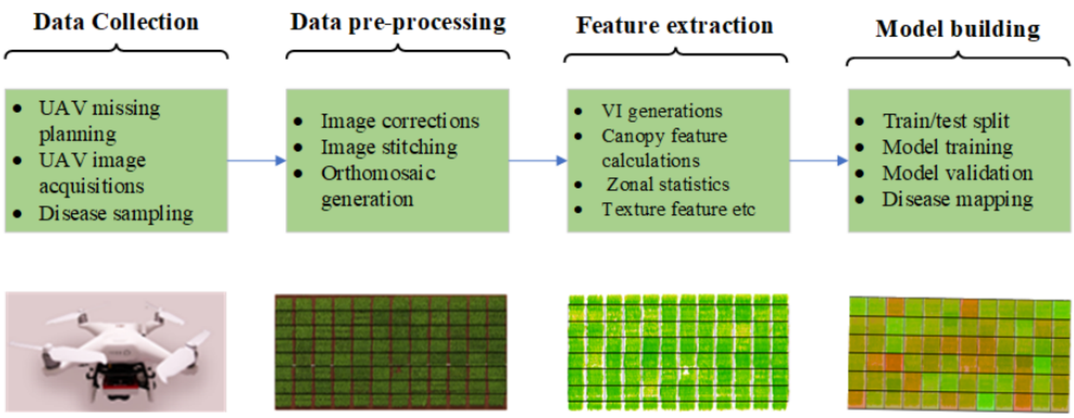

Figure 3.12: General Workflow of Conventional ML-Based Crop Disease Detection Using UAV Imagery (Shahi et al., 2023).

Researchers have applied various machine learning (ML) methods to detect wheat diseases, with different results depending on the disease type and the input features used. Here is a detailed explanation of the studies:

- **(Liu et al., 2020)** worked on detecting Fusarium head blight using hyperspectral sensors. They utilized spectral bands, vegetation indices, and texture features as input features. They applied a Backpropagation neural network with Simulated Annealing, a technique that helps optimize the model by avoiding local minima, and achieved a high accuracy of 98%.
- **(Ma et al., 2021)** focused on detecting Fusarium head blight using hyperspectral sensors. They used spectral bands, vegetation indices, and wavelet features. The machine learning

method employed was Support Vector Machine (SVM), which yielded a coefficient of determination (R^2) of 0.88, indicating a strong predictive performance.

- **(Zhu et al., 2022)** worked on detecting wheat scab using multispectral sensors. They used vegetation indices and texture features as input features. The models used for analysis included Partial Least Squares Regression (PLSR), Support Vector Regression (SVR), and Backpropagation Neural Networks (backpropagation neural network (BPNN)), achieving an R^2 value of 0.83.

- **(Bohnenkamp et al., 2019)** focused on detecting yellow rust using hyperspectral sensors, with vegetation indices as the primary input feature. They applied the Support Vector Machine (SVM) method and obtained an R^2 value of 0.63.

Table 3.3 provides a summary of conventional machine learning methods for wheat disease estimation, highlighting the combination of sensor types, features, and machine learning techniques employed to improve disease detection accuracy.

Table 3.3: Summary of conventional ML methods for wheat disease estimation.

Reference	**Disease**	**Sensors**	**Features**	**ML Methods**	**Eval. Metrics**
(Liu et al., 2020)	Fusarium head blight	HS	SBs (spectral bands), VI, TF (texture features)	BP with SA	Accuracy = 0.98
(Ma et al., 2021)	Fusarium head blight	HS	SBs (spectral bands), VIs, WFs	SVM	$R^2 = 0.88$
(Zhu et al., 2022)	Wheat scab	MS	VI, TF (texture features)	PLSR, SVR, BPNN	$R^2 = 0.83$
(Bohnenkamp et al., 2019)	Yellow rust	HS	VIs	SVM	$R^2 = 0.63$

3.6.4 Deep learning (DL)-Based methods

Deep learning methods have been widely applied for crop disease estimation using UAV imagery. The general pipeline for deep learning-based crop disease detection is shown in Figure 3.13 and includes data collection, preparation, model building, and evaluation. However, specific tasks such as image stitching, tiling, and annotation are critical during data preparation for UAV images. Deep learning models for crop disease detection can be categorized into classification-based, segmentation-based, and detection-based approaches. Segmentation models classify individual pixels as healthy or diseased, while classification models classify entire images into disease categories (Shahi et al., 2023). Detection-based models localize and label the area of interest within the image, as illustrated in Figure 3.13.

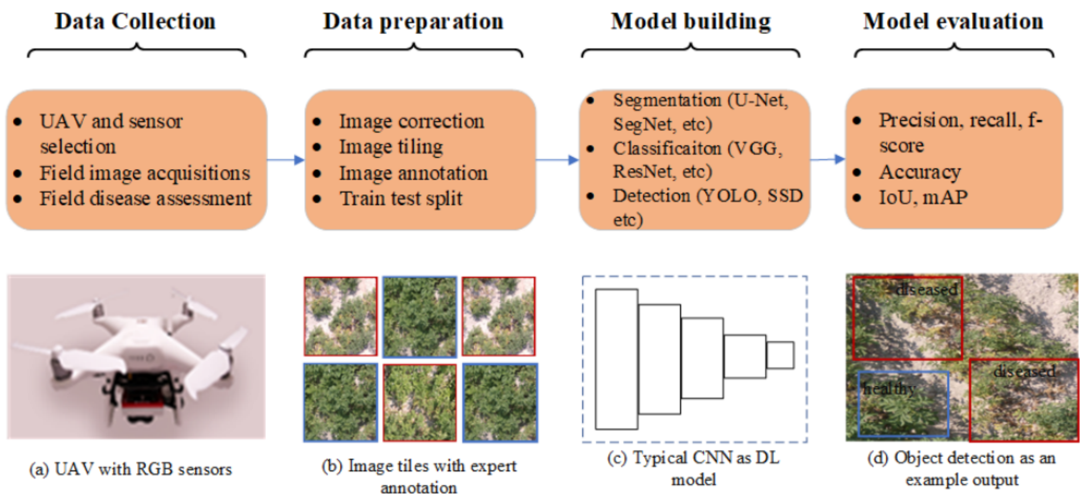

Figure 3.13: General Workflow of DL-based crop disease detection using UAV imagery (Shahi et al., 2023).

Pixel-Based Segmentation Models

Pixel-based segmentation models focus on classifying each pixel in an image based on its characteristics, allowing for precise detection of disease symptoms in specific areas of a plant or field. By analyzing the image at a granular level, these models assign labels to each pixel, helping to identify localized disease patterns across the crop. Several studies have applied such models to detect wheat diseases, yielding impressive results. For instance, (Pan et al., 2021) used RGB sensors and the PSPNet model to detect yellow rust, achieving an accuracy of 94%. Similarly, (Su et al., 2020) applied multispectral sensors and the U-Net model for yellow rust, achieving a recall of 0.926 and an F-score of 0.92. (Deng et al., 2022) employed RGB sensors with the DeepLabv3+ model for stem rust, obtaining an F-score of 0.81. (T. Zhang et al., 2021) used RGB sensors and the lr-UNet model for yellow rust, achieving an accuracy of 97.13%, while (T. Zhang et al., 2022) applied multispectral sensors with the UNet and DF-UNet models for yellow rust, achieving an accuracy of 96.93%. These studies demonstrate the effectiveness of pixel-based segmentation models in detecting wheat diseases with high precision.

Object-Level classification models

Object-level classification models treat the image as a whole, classifying entire regions or objects within an image rather than individual pixels. These models focus on detecting the overall presence of diseases at the object or plant level, which is useful for broader assessments of crop health. For instance, (X. Zhang et al., 2019) used hyperspectral sensors and the Inception-ResNet model for detecting yellow rust, achieving an accuracy of 85%. Similarly, (Huang et al., 2019) applied RGB sensors and a convolutional neural network (CNN) to detect Helminthosporium leaf blotch, achieving an accuracy of 91.43%. These studies highlight the application of object-level classification models in detecting crop diseases at a larger scale, emphasizing overall disease presence across plant regions.

A summary of deep learning-based methods for wheat disease estimation is presented in Table 3.4. Overall, these deep learning approaches show great potential for fine-grained and scalable wheat disease monitoring in precision agriculture.

Table 3.4: Summary of deep learning-based methods for wheat disease estimation.

Reference	Disease	Sensors	Task	DL Methods	Results
(Pan et al., 2021)	Yellow Rust	RGB	Pixel-Based Segmentation Models	PSPNet	Acc = 0.94
(Su et al., 2020)	Yellow Rust	MS	Pixel-Based Segmentation Models	U-Net	Recall = 0.926, F-Score = 0.92
(Deng et al., 2022)	Stem Rust	RGB	Pixel-Based Segmentation Models	DeepLabv3+	F-Score = 0.81
(T. Zhang et al., 2021)	Yellow Rust	RGB	Pixel-Based Segmentation Models	lr-UNet	Acc = 0.9713
(T. Zhang et al., 2022)	Yellow Rust	MS	Pixel-Based Segmentation Models	UNet, DF-UNet	Acc = 96.93
(X. Zhang et al., 2019)	Yellow Rust	HS	Object-Level Classification Models	Inception-ResNet	Acc = 0.85
(Huang et al., 2019)	Helminthosporium leaf blotch	RGB	Object-Level Classification Models	CNN	Acc = 0.9143

RGB sensors are predominantly paired with deep learning (DL) methods, while MS sensors are more frequently used with machine learning (ML) approaches. This distinction is closely tied to the spatial resolution requirements of DL models, which demand high-resolution images typically captured at lower altitudes using RGB sensors. In contrast, ML models are more tolerant of lower-resolution imagery, often obtained from higher flight altitudes. As illustrated in Figure 10, most DL-based studies used UAV images taken below 20 meters, whereas ML-based studies tended to operate UAVs above 30 meters. The limited use of hyperspectral sensors in both ML and DL methods is likely due to their high cost and the complexity of data processing. These trends underscore the growing preference for cost-effective, high-resolution UAV setups in DL-based crop disease estimation (Shahi et al., 2023).

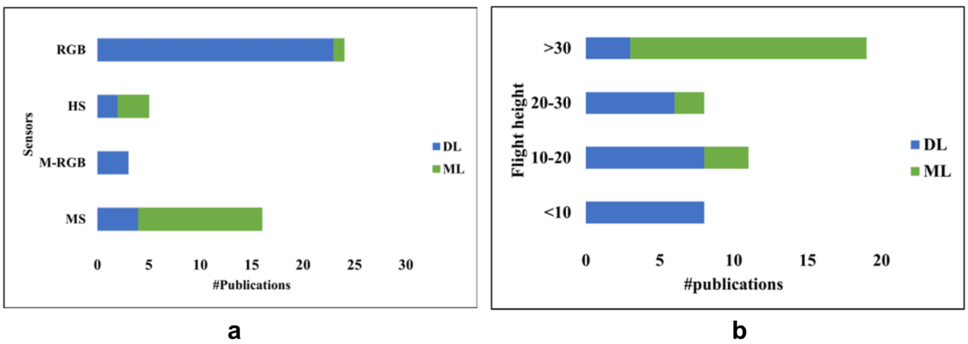

Figure 3.14: The distribution of existing works based on (a) sensors and (b) flight altitudes used in UAV image acquisition. Note that M-RGB denotes modified RGB sensors (Shahi et al., 2023).

3.7 Fusion of satellite and UAV data

The fusion of satellite and UAV data presents a promising solution for overcoming the limitations of each platform in agricultural monitoring. While satellites offer broad and frequent coverage, their spatial resolution can be insufficient for detailed field assessments. UAVs provide high-resolution imagery but are limited in coverage and frequency. By combining these complementary strengths, it becomes possible to achieve accurate, timely, and scalable crop monitoring that supports more informed decision-making in precision agriculture (Y. Li et al., 2022; Maimaitijiang et al., 2020).

3.7.1 Systematic categorization of UAV/Satellite monitoring methods

UAV/Satellite integration is categorized using a hierarchical decision tree based on three criteria (Figure 3.15). The first distinguishes between weak and strong synergies: Weak synergy involves simple comparisons of UAV and satellite data, while strong synergy combines them to produce more informative outcomes. Among strong synergies, the second criterion differentiates between cases with the same observation target and those with different scales, known as "multiscale explanation," where UAV data refines or contextualizes broader satellite observations. The third criterion identifies whether models are calibrated using one or both data sources, referred to as "model calibration" (using one to support the other) or "data fusion" (using both to generate new, enhanced data). Data fusion techniques are further divided into pixel-level, feature-level, and decision-level fusion, depending on the depth of integration. Precision agriculture applies all types of synergy strategies, with a slight preference for data fusion. This suggests that precision agriculture benefits from both the spatial detail of UAVs and the broader coverage of satellite data to enhance monitoring and decision-making (Alvarez-Vanhard et al., 2021).

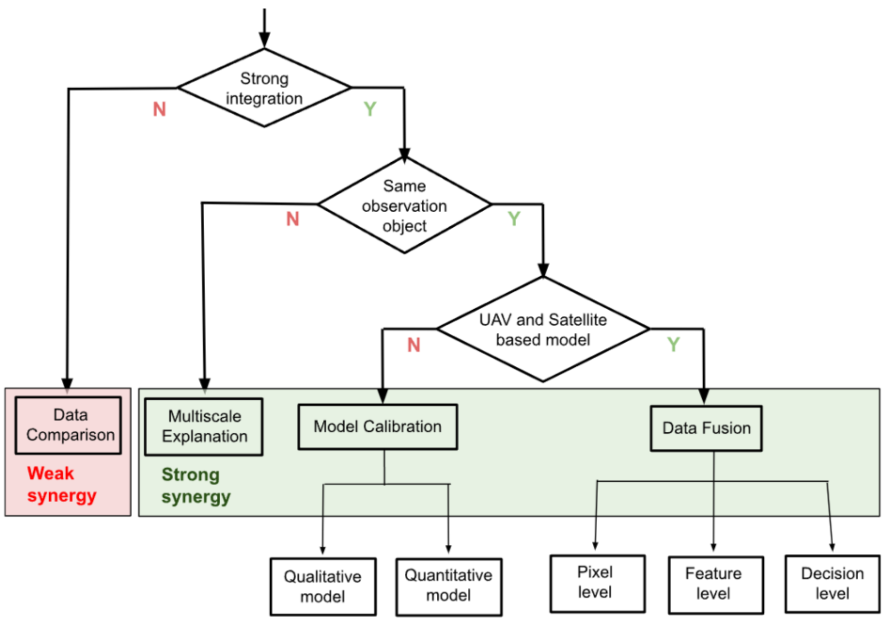

Figure 3.15: Hierarchical decision tree for categorizing UAV/Satellite strategies (Alvarez-Vanhard et al., 2021).

Data comparison strategy

The data comparison strategy represents a weak form of UAV/Satellite synergy, where data from both sources are analyzed separately rather than combined (Figure 3.16). These studies highlight the complementary strengths of each platform: satellites offer broad coverage and standardized processing, which is useful for monitoring large or inaccessible areas, while UAVs provide very high spatial resolution at low cost, which is ideal for capturing fine details like crop variability or small water bodies. UAVs are especially valuable in precision agriculture due to their flexibility in capturing data at critical times for input decisions. Ultimately, the choice between UAV and satellite data depends on the study's scale, goals, and available resources. Notably, 19% of these comparison studies acknowledged the potential for stronger synergies in the future (Alvarez-Vanhard et al., 2021).

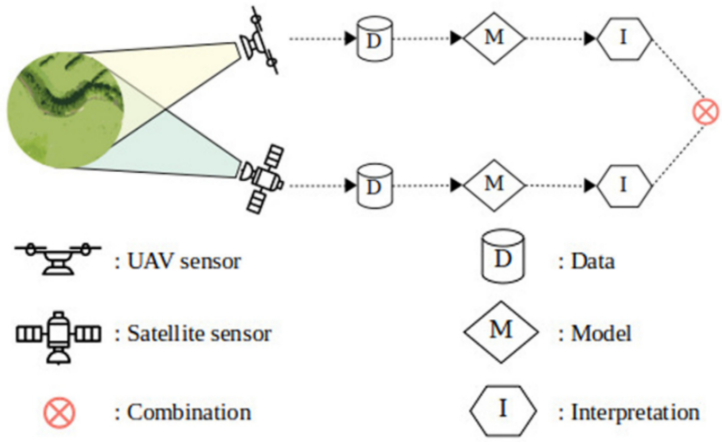

Figure 3.16: Diagram of data comparison strategy (Alvarez-Vanhard et al., 2021).

Multiscale explanation strategy

The multiscale explanation strategy leverages the different spatial scales of UAV and satellite data to enhance interpretation (Figure 3.17). Satellites provide a broad, regional context, while UAVs offer detailed, fine-scale information. This strategy enables better understanding by combining wide-area observation with localized, high-resolution insights, often using UAV-derived digital surface models to refine spatial analysis (Alvarez-Vanhard et al., 2021).

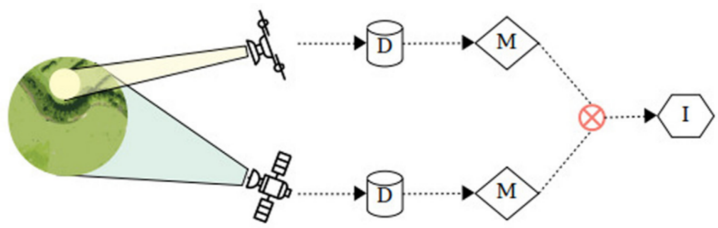

Figure 3.17: Diagram of the Multiscale explanation strategy (Alvarez-Vanhard et al., 2021).

Model calibration strategy

The model calibration strategy involves using one data source, often UAV or satellite imagery, to calibrate models built on the other (Figure 3.18). It was the most frequently used strong synergy approach, especially in ecology and precision agriculture. This strategy can be qualitative, where UAV-derived labels (from expert interpretation, thresholds, or classifications) train satellite-based classification models, or quantitative, where UAV-derived numerical values (e.g., chlorophyll, biomass, height) calibrate regression or unmixing models. In some cases, UAV data replaces traditional ground surveys, serving as the sole reference source. While in-situ data is still used in many studies for validation, UAVs are increasingly relied upon for producing accurate, high-resolution ground truth. The strategy also includes data intercalibration, where one dataset is used to standardize or refine the spectral characteristics of the other (Alvarez-Vanhard et al., 2021).

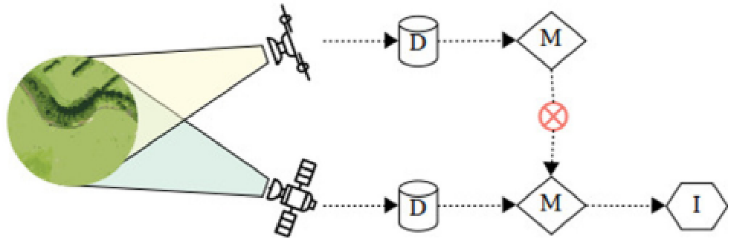

Figure 3.18: Diagram of Model calibration strategy (Alvarez-Vanhard et al., 2021).

Data fusion strategy

The data fusion strategy represents the strongest form of UAV/Satellite synergy, aiming to fully integrate both data sources to create enhanced datasets (Figure 3.19). Though less commonly used, it has been explored in precision agriculture for extracting fine-resolution land cover and vegetation traits. Most studies applied pixel-level fusion to combine spatial, spectral, or temporal details, improving classification accuracy, resolution, or time-series completeness. Some studies used feature-level fusion, integrating complementary views from UAVs and satellites for tasks like damage assessment or change detection. While decision-level fusion holds potential, it has not yet been applied in this context. Overall, data fusion enables more accurate and consistent monitoring by leveraging the unique strengths of both platforms (Alvarez-Vanhard et al., 2021).

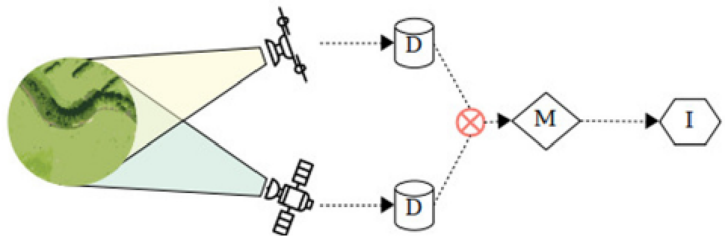

Figure 3.19: Diagram of Data fusion strategy (Alvarez-Vanhard et al., 2021).

3.7.2 Challenges of UAV/Satellite data fusion

While combining UAV and satellite data offers great potential, it also presents several challenges, including:

- **Limited Exploitation of Synergy:** The potential of UAV/Satellite synergy remains underexploited, as most studies prioritize validation or simple comparisons rather than full integration; advanced strategies like multiscale explanation and data fusion, which preserve and leverage the strengths of both data sources, are still rarely applied (Alvarez-Vanhard et al., 2021).
- **Interoperability Issues:** Interoperability is a major challenge due to the lack of standardization in UAV data acquisition and quality assurance, unlike standardized satellite

data; variations in sensors, protocols, and conditions further complicate the alignment of multi-source datasets (Alvarez-Vanhard et al., 2021).

- **Geometric and Radiometric Misalignment:** Geometric and radiometric misalignment pose a challenge, as misregistration between UAV and satellite data impacts multi-scale analysis; sub-pixel geometric calibration is difficult due to resolution gaps, and radiometric inconsistencies from sensor and environmental differences further hinder seamless integration (Alvarez-Vanhard et al., 2021).

- **Technical Complexity:** Fusion of UAV and satellite data remains technically complex, as advanced methods like spatio-temporal and spectral-temporal fusion require specialized expertise, making them less accessible and limiting wider adoption (Alvarez-Vanhard et al., 2021).

3.8 Discussion

After an in-depth exploration of the literature, we have identified that the integration of AI-based models especially machine learning (ML) and deep learning (DL) with remote sensing represents a promising approach for early wheat disease detection. However, this integration is far from straightforward and presents several key challenges that emerged during our research.

One of the most significant obstacles is the lack of annotated datasets that combine remote sensing imagery (e.g., UAV, satellite) with accurate disease labels. Deep learning models require large amounts of labeled data to generalize effectively, yet in agricultural settings, acquiring such datasets is both time-consuming and costly. This shortage limits the performance and scalability of AI models across different regions and disease types.

Additionally, the computational complexity of deep learning algorithms poses a barrier to their use in real-world field conditions. Most of the models analyzed in the reviewed studies require powerful GPUs and substantial memory, making them unsuitable for real-time processing on drones or mobile devices, particularly in resource-constrained environments.

Another challenge is the context sensitivity of model performance. The success of AI-based disease detection systems is influenced by several factors, including sensor type, flight altitude, weather conditions, crop growth stages, and disease severity. This variability complicates the development of models that are robust and transferable across diverse agricultural scenarios.

Despite these limitations, the integration of AI and remote sensing continues to offer compelling opportunities. Based on insights drawn from the state of the art, we believe that the development of lightweight, edge-compatible models will be essential for enabling real-time disease detection on UAVs and other portable platforms. Such models would allow farmers to receive immediate feedback without relying on cloud infrastructure.

Moreover, the fusion of data from multiple sensing platforms for example, combining satellite, UAV, and ground-based data can enhance the spatial and spectral resolution available to AI models, resulting in more accurate and reliable detection outcomes.

To reduce reliance on large annotated datasets, semi-supervised learning and transfer learning are emerging as viable strategies. These techniques allow AI models to learn effectively from limited labeled data by leveraging prior knowledge or incorporating abundant unlabeled data.

Finally, we observed a growing need for explainable AI systems in agriculture. For AI-driven remote sensing tools to gain trust and wide adoption among farmers, it is essential that the models not only generate predictions but also provide interpretable explanations that users can understand and act upon.

In summary, while the integration of AI with remote sensing for wheat disease detection is still evolving, our review confirms its strong potential to transform agricultural monitoring. Addressing current limitations through lightweight modeling, data fusion, and explainability will be crucial to developing effective, accessible, and farmer-centered solutions.

3.9 Conclusion

The integration of remote sensing with machine learning and deep learning has opened new possibilities for the efficient and accurate detection of wheat diseases. By leveraging data from satellites, UAVs, and other sensing platforms, combined with advanced learning algorithms, it is now possible to monitor crop health at scale, detect early signs of infection, and support timely intervention. This chapter highlighted the main technologies, workflows, and challenges involved in this integration, emphasizing its value in enhancing disease management and promoting sustainable agricultural practices. Continued research and refinement of these methods will be essential to fully realize their potential in real-world farming environments.

General Conclusion

The increasing global demand for wheat, coupled with the persistent threat of plant diseases and insect pests, underscores the urgent need for innovative and efficient monitoring systems in agriculture. Traditional disease detection methods, while useful, are often limited in scale, accuracy, and timeliness. This has prompted the adoption of smart agriculture technologies, particularly the integration of remote sensing with machine learning (ML) and deep learning (DL) techniques.

Throughout this report, we have examined how the fusion of aerial imagery acquired through satellites, UAVs, and multispectral sensors with advanced AI models enables early detection, classification, and monitoring of various wheat diseases. We have explored the fundamentals of DL and ML, including their roles in core computer vision tasks, and reviewed different imaging technologies used in precision agriculture.

Moreover, we discussed the challenges inherent in this integration, such as data heterogeneity, high computational demands, and the need for large annotated datasets. Despite these limitations, the results from existing studies demonstrate that remote sensing combined with AI holds immense potential for automated, scalable, and real-time disease detection systems.

In conclusion, the integration of remote sensing and intelligent algorithms is not only transforming the way we monitor crop health but also paving the way for sustainable, data-driven agricultural practices. Future work should focus on improving data fusion strategies, model generalization, and deploying lightweight solutions for field-level implementation bringing us closer to the vision of fully autonomous smart farming systems.

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Appendix A

Overview of Lightweight Neural Network Architectures and Vegetation Indices for Plant Disease Detection

1.1 CNN Architectures

This section presents several notable Convolutional Neural Network (CNN) architectures that have significantly advanced image recognition tasks, particularly in the context of plant disease detection and classification. Each architecture introduces unique design principles to improve accuracy, efficiency, and scalability.

Visual geometry group network (VGGNet)

Proposed by Simonyan and Zisserman, VGGNet is a convolutional neural network (CNN) architecture widely recognized for its simplicity and strong performance in image recognition tasks (Alzubaidi et al., 2021).

VGG is characterized by its deep architecture, typically comprising 16 to 19 layers. One of its key innovations is the replacement of large convolutional filters (such as 11×11 or 5×5) with multiple stacked 3×3 filters. This strategy maintains an equivalent receptive field while reducing the number of parameters and improving computational efficiency.

In addition, VGG uses 1×1 convolutions to control the model's complexity and includes max pooling layers to progressively reduce the spatial dimensions of the feature maps, as illustrated in figure 1.1.Despite its effectiveness, a major drawback of VGGNet is its high computational cost, with around 140 million parameters (Alzubaidi et al., 2021).

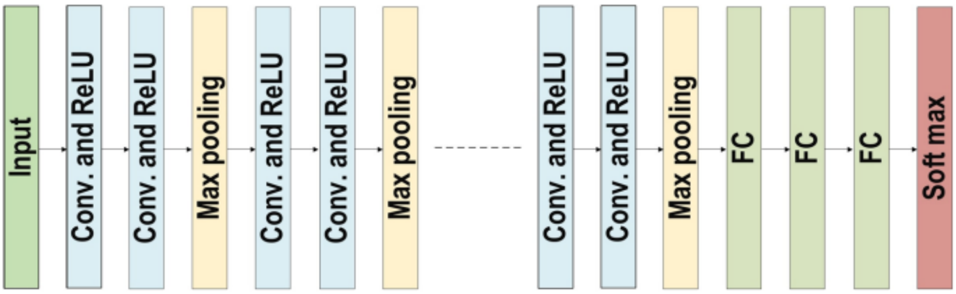

Figure 1.1: The architecture of VGGNet (Alzubaidi et al., 2021).

Inception Net (GoogLeNet)

Inception Net, introduced by Szegedy et al., uses Inception modules that apply multiple convolutional filters (1×1, 3×3, 5×5) in parallel, followed by concatenation (see Figure 1.2). This design captures multi-scale information efficiently while reducing computational cost. The architecture has been successfully applied in plant phenotyping and classification of complex disease patterns.

It replaced standard convolutional layers with micro-neural networks and regulated computation through 1×1 convolutions as bottleneck layers. Sparse connections addressed redundant information by selectively connecting input and output channels, while the global average pooling (GAP) layer reduced parameters from 40 million to just 5 million, enhancing efficiency. Additional features included the RmsProp optimizer, batch normalization, and auxiliary learners to accelerate convergence (Alzubaidi et al., 2021).

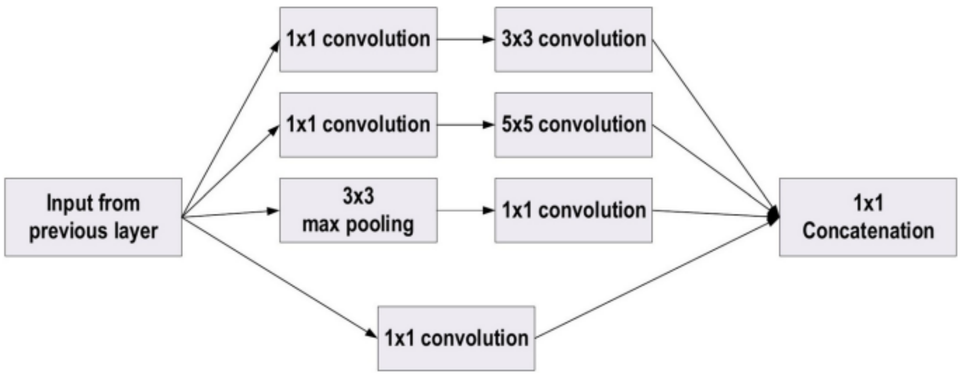

Figure 1.2: The basic structure of Google Block (Alzubaidi et al., 2021).

Xception Net

The Xception model is an extension of the Inception architecture that replaces standard convolutions with depth-wise separable convolutions, significantly improving efficiency. It has been shown to outperform Inception in many tasks, especially with high-resolution agricultural images, by learning spatial and cross-channel correlations separately.

The core concept behind Xception is the modification of the traditional Inception block by making it wider and replacing the standard 3×3 convolution followed by a 1×1 convolution with depthwise separable convolutions, which reduces computational complexity while enhancing performance (Alzubaidi et al., 2021). An illustration of the basic Xception block structure is presented in Figure 1.3.

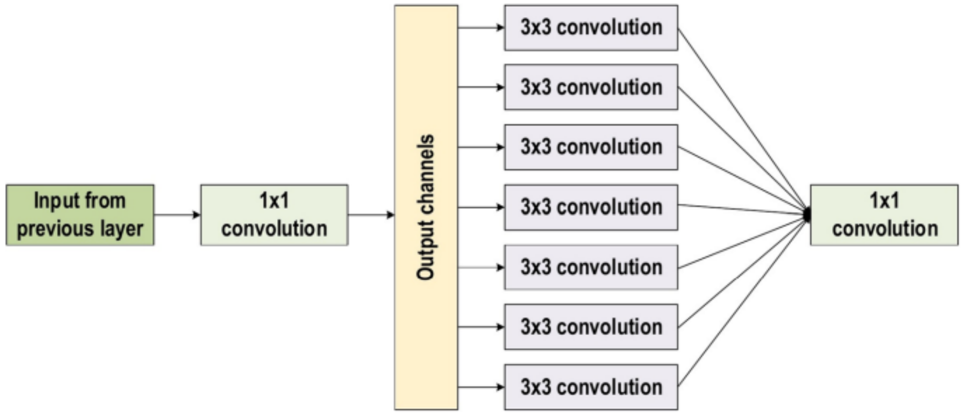

Figure 1.3: The basic block diagram for the Xception block architecture (Alzubaidi et al., 2021).

Residual Networks (ResNet)

ResNet is a deep convolutional neural network architecture developed to facilitate the training of very deep networks, ranging from 18 to 152 layers, by introducing the concept of residual learning through shortcut (or skip) connections that bypass one or more layers.

These residual connections address the degradation problem that often occurs in deeper networks, where increasing depth leads to performance saturation or even degradation. By enabling gradients to flow more efficiently, ResNet makes it easier for the network to learn identity mappings or residual functions, simplifying the overall training process.

The output of a residual block is defined as:

$$H(x) = F(x) + x \tag{3}$$

Where:

- $H(x)$: the output of the residual block,
- $F(x)$: the residual function to be learned,
- x: the input passed through the shortcut connection.

This formulation (3) makes it easier to optimize deep networks by learning the difference (residual) between the desired mapping and the identity. Instead of learning the full transformation $H(x)$, the network focuses on learning the residual function $F(x) = H(x) - x$, which is often easier to optimize, especially when the identity mapping is a reasonable starting point. This approach improves gradient flow during training, leading to better convergence and more stable deep models.

An illustration of the residual module structure is provided in Figure 1.4.

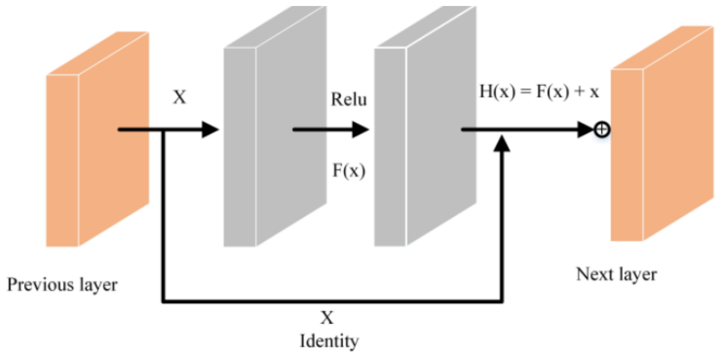

Figure 1.4: Residual module diagram (Fang et al., 2023).

DenseNet

DenseNet is a convolutional neural network architecture that introduces the concept of dense connectivity, in which each layer is directly connected to every other layer in a feed-forward manner, as illustrated in Figure 1.5. This unique design enables feature reuse, enhances gradient flow, and significantly reduces the number of parameters compared to traditional CNNs.

Inspired by ResNet and Highway Networks, DenseNet addresses a key limitation in ResNet, where each layer maintains isolated weights and where certain transformations contribute minimal new information. In contrast, DenseNet concatenates the outputs of all preceding layers and feeds them as input to each subsequent layer.

In a DenseNet with l layers, the number of direct connections between layers is given by:

$$\text{Number of connections} = \frac{l(l+1)}{2} \tag{4}$$

This connectivity pattern facilitates richer feature propagation, introduces a regularization effect, and helps mitigate the vanishing gradient problem, thus improving training efficiency and model generalization (Alzubaidi et al., 2021).

Despite the computational cost introduced by the accumulation of feature maps, DenseNet has shown exceptional performance in fine-grained classification tasks, such as distinguishing between subtly different plant disease symptoms, particularly in scenarios with limited training data.

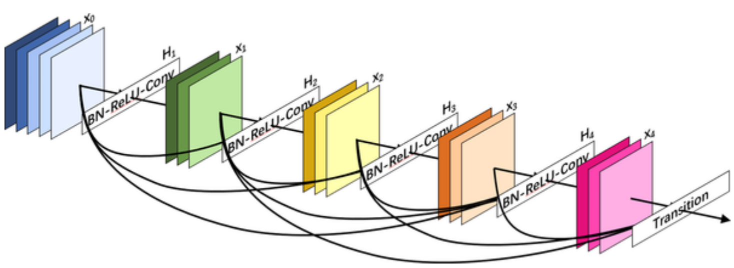

Figure 1.5: The architecture of DenseNet Network (BN is batch normalisation) (Alzubaidi et al., 2021).

1.2 Lightweight Neural Network Architectures

Lightweight neural network architectures are designed to be efficient, enabling high performance on resource-constrained devices like mobile phones and edge devices. These models balance accuracy and computational efficiency, making them ideal for real-time applications. The following are some key lightweight architectures and how they address these challenges.

1.2.1 MobileNetV1

MobileNet V1 is a lightweight convolutional neural network that efficiently processes input images by utilizing depthwise separable convolutions to reduce computational cost and model size. Starting with an input image resized to 224×224×3, the network first applies a standard convolution followed by a series of depthwise separable convolutions that decompose standard convolutions into two steps: a depthwise convolution that filters each input channel independently and a pointwise (1×1) convolution that combines these filtered outputs. This approach drastically reduces the number of parameters and operations, as seen in the progressive reduction of feature map dimensions from 112×112×32 to 56×56×64, 28×28×256, and eventually to 7×7×1024 (Figure 1.6). An average pooling operation then reduces the spatial dimension to 1×1×1024, after which a fully connected layer performs the final classification (Kadam et al., 2022).

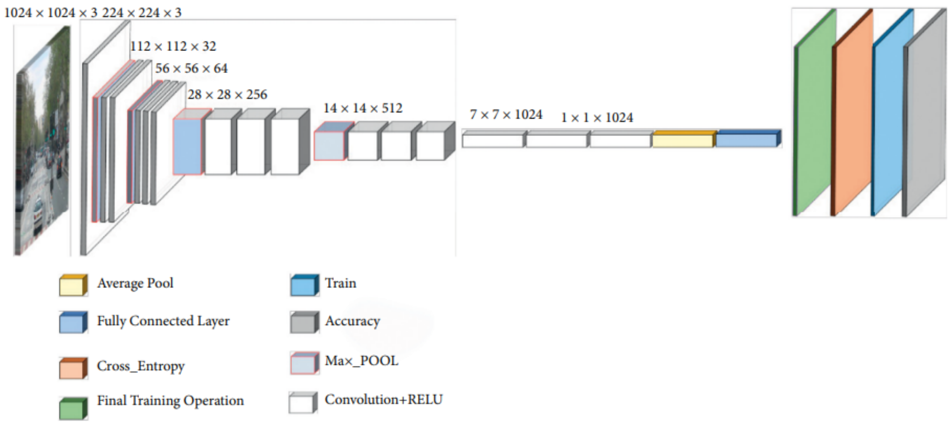

Figure 1.6: The architecture of MobileNet V1 (Kadam et al., 2022).

1.2.2 EfficientNet

EfficientNet is a family of convolutional neural networks developed by Google AI that introduces a compound scaling method to efficiently scale deep learning models. Traditional approaches often scale models arbitrarily in one of three dimensions: depth (number of layers), width (number of channels), or input resolution. However, EfficientNet proposes a more balanced and systematic strategy, where all three dimensions are scaled simultaneously and proportionally using a fixed set of scaling coefficients. This compound approach maintains model efficiency while significantly boosting accuracy.

The baseline model, EfficientNet-B0 (see Figure 1.7), is built using Neural Architecture Search (NAS) to optimize both performance and efficiency. Larger variants (B1 to B7) are derived by uniformly scaling the baseline model using the compound scaling principle. This results in models that achieve state-of-the-art performance on image classification tasks with dramatically fewer parameters and lower computational cost compared to earlier architectures like ResNet or Inception.

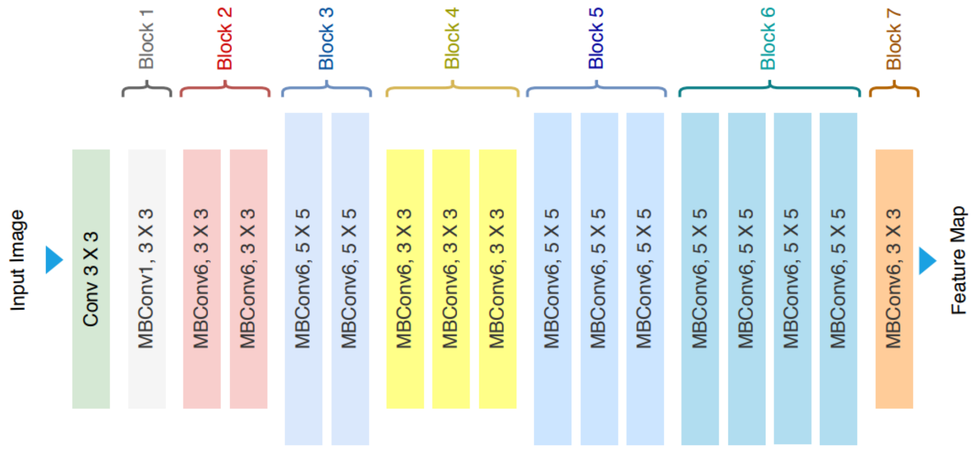

Figure 1.7: Architecture of EfficientNet-B0 (Ahmed & Sabab, 2022).

1.2.3 ShuffleNet V2

ShuffleNet V2 is a family of lightweight convolutional neural networks specifically designed for mobile and embedded platforms with limited computational resources. It improves upon ShuffleNet V1 by addressing practical deployment issues, such as memory access cost and actual inference latency, which are often overlooked in models optimized purely for FLOPs.

Unlike previous versions that relied heavily on group convolutions, ShuffleNet V2 introduces a simpler and more efficient building block. This block incorporates a channel split operation, followed by pointwise and depthwise convolutions, and concludes with a channel shuffle operation. By delaying the channel shuffle to the end of the split-transform-merge sequence, the model enhances feature mixing between channel groups and improves accuracy without incurring a significant computational burden.

These architectural improvements allow ShuffleNet V2 to achieve a strong trade-off between speed and accuracy, making it a highly practical choice for real-time applications in edge computing and mobile vision systems.

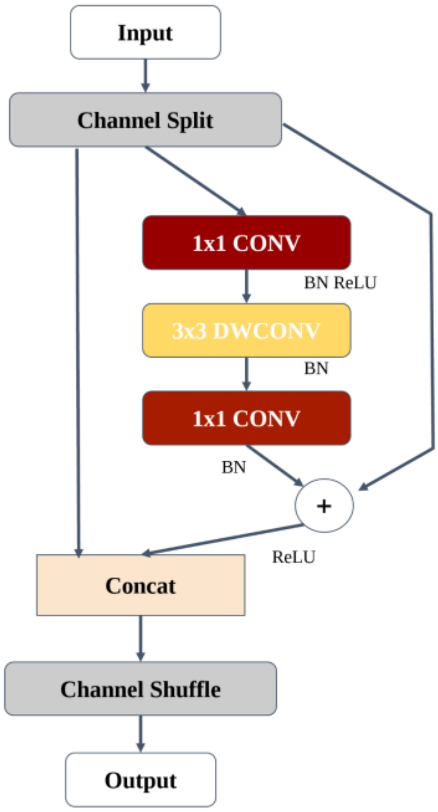

Figure 1.8: Architecture of the basic building block of ShuffleNet V2 with residual connection. CONV: convolution layer; DWCONV: depthwise convolution; BN: batch normalization. Channel Shuffle: a key operation enabling inter-group communication in ShuffleNet architectures (Dobko et al., 2020).

1.3 Vegitation Indices

The table below (Table 1.1) summarizes the vegetation indices used in the previously discussed studies, along with their formulas and applications.

Table 1.1: Vegetation Indices and Their Applications (where R = Red, G = Green, B = Blue, and NIR = Near Infrared)

Reference	Index	Formula	Application
(Bhandari et al., 2020)	Green Leaf Index (GLI)	$\mathrm{GLI} = \frac{2\times G-R-B}{2\times G+R+B}$	Detect green canopy cover in wheat.
(Bhandari et al., 2020)	Green Index (GI)	$\mathrm{GI} = \frac{G}{R}$	Detecting disease stress in crops.
(Bhandari et al., 2020)	Normalized Difference Index (NDI)	$\mathrm{NDI} = \frac{G-R}{G+R}$	Separates plants from soil in RGB images.
(Heidarian Dehkordi et al., 2020)	Stripe Rust Index (SRI)	$\mathrm{SRI} = R+G$	Detects wheat stripe rust infection by identifying yellow-colored regions.
(Heidarian Dehkordi et al., 2020)	Leaf Rust Index (LRI)	$\mathrm{LRI} = \frac{2\times G-R^2}{R\times(G-B)}$	Identifies wheat leaf rust based on brown color codes by comparing dark and light brown spectral compositions derived from RGB bands.
(Guo et al., 2021)	SIPI	$\mathrm{SIPI} = \frac{NIR-B}{NIR-R}$	Detects carotenoid-to-chlorophyll ratio; useful in early plant stress and senescence monitoring.
(Guo et al., 2021)	PRI	$\mathrm{PRI} = \frac{G-B}{G+B}$	Indicates photosynthetic light use efficiency, sensitive to physiological changes before visible symptoms.
(Guo et al., 2021)	TCARI	$\mathrm{TCARI} = 3\times[(R-G)-0.2\times(R-B)]\times\left(\frac{R}{G}\right)$	Estimates chlorophyll content, compensating for soil background, useful in chlorosis or disease stress.
(Guo et al., 2021)	PSRI	$\mathrm{PSRI} = \frac{R-B}{NIR}$	Indicates leaf senescence and pigment degradation, useful for detecting aging or stressed vegetation.
(Guo et al., 2021)	YRI	$\mathrm{YRI} = \frac{NIR-B}{NIR+B-0.5\times R}$	Developed to identify yellow rust in wheat; captures yellowing due to fungal infection.
(Guo et al., 2021)	MSR	$\mathrm{MSR} = \frac{\frac{NIR}{R}-1}{\frac{NIR}{R}+1}$	Enhances contrast in vegetation vigor; reduces soil effects; used in disease detection (e.g., powdery mildew).
(Guo et al., 2021)	MCARI	$\mathrm{MCARI} = [(G-R)-0.2\times(G-B)]\times\left(\frac{G}{R}\right)$	Highlights chlorophyll absorption, useful in chlorophyll estimation and early stress detection.
(Guo et al., 2021)	Powdery Mildew Index (PMI)	$\mathrm{PMI} = \frac{G-R}{G+R-0.5\times NIR}$	Identifies powdery mildew by sensing spectral changes in plant pigments and moisture.